MAHARASHTRA STATE BOARD OF TECHNICAL EDUCATION, MUMBAI

A Project Report on

Digital GroceryHub System

Submitted as a partial fulfillment for the award of

Diploma in Computer Engineering

Submitted by

Baisane Narendra Prabhakar

Bhavsar Satyam Paresh

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DEPARTMENT OF COMPUTER ENGINEERING

S.S.V.P.S's Bapusaheb Shivajirao Deore Polytechnic, Dhule

(Institute Code : 0059)

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CERTIFICATE

This is to certify that,

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has satisfactorily completed project entitled “Digital GroceryHub System” in partial fulfillment for the **Diploma Course in Computer Engineering** of the **Maharashtra State Board of Technical Education** at our polytechnic during the academic year **2025-2026**. The project is completed by a group consisting of four persons under the guidance of the Faculty Guide.

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Principal.

Acknowledgment

We would like to express our sincerest gratitude to all people who have contributed towards our project. We would like to extent our heartiest thanks to the Head of Computer Department, our mentor Ms. M. L. Manikhedkar mam and subject teacher Mr. N. D. Patel sir for their important guidance, help and advice which was really important for our overall capstone project.

Also, we would like to extend our thanks to all the faculty members of the Department of Computer Engineering and to all our colleagues who helped us in completing this project.

Baisane Narendra Prabhakar
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Abstract

The proposed system is “Digital GroceryHub System”. It is a web application. In today’s fast lifestyle, people face difficulty in managing grocery shopping due to time constraints and lack of proper planning. This system is like a digital helper designed for customers and shop owners, by creating an environment to manage grocery activities efficiently. The before System makes it easy to browse, order, and manage grocery products. The primary goal is to streamline the entire process of grocery management, from product selection to delivery. It aims to simplify the process and reduce manual work. The system provides tracking of orders, allowing customers and admin to track stages including product selection, order placement, payment, and delivery. System encourages proper documentation of transactions and product details, making it easy to retrieve when needed. System provides a user-friendly interface to ensure that users and administrators can easily navigate and manage the system efficiently.

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Chapter 1

Introduction and Background of Industry

1.1 Introduction:

The Digital GroceryHub System is designed to make grocery shopping simple and convenient in today's busy lifestyle. It allows users to purchase daily essentials online without spending time in physical stores. The system provides a smooth digital shopping experience where users can easily browse and order products from anywhere, helping them save time and effort.

The platform offers a wide range of products organized into categories such as grains, pulses, oils, spices, snacks, dairy items, and household essentials. It focuses on providing high-quality dry goods and packaged products while ensuring that items are always available. The system also helps users quickly find what they need through a user-friendly interface.

The Digital GroceryHub System also ensures efficient order processing and delivery. Once an order is placed, products are carefully packed and delivered on time. With a secure checkout process and reliable service, the system aims to provide a convenient, affordable, and modern solution for everyday grocery shopping.

Additionally, the system maintains proper records of user activities, orders, and product details, which helps in better management and tracking. It supports secure user authentication and ensures that only authorized users can access the system. Overall, the system improves efficiency, reduces manual work, and enhances the overall grocery shopping experience for users.

1.2 Background of the Industry:

According to the rapid growth of digital technology and increasing use of the internet, online systems have become an important part of daily life. In today's fast-paced world, people prefer digital solutions that save time and effort, and the grocery industry is also shifting from traditional shopping methods to online platforms. The Digital GroceryHub System is developed to meet this need by providing a simple and efficient way to manage grocery shopping through a web application. An online grocery system may be defined as a digital platform that allows users to select, order, and manage grocery items efficiently

by integrating functionalities such as product management, order processing, and delivery tracking. Such systems improve convenience, reduce manual work, ensure proper record maintenance, and provide a user-friendly interface, making grocery shopping faster, easier, and more efficient.

a) Easy access to grocery products anytime and anywhere

b) Time-saving and convenience for users

c) Efficient management of orders and delivery

d) Proper organization of products and inventory

e) User-friendly interface for better experience

f) Reduction in manual work and errors

g) Proper record maintenance and easy retrieval of data

Capstone Project

One of the meanings of a capstone is the final stage that completes a process, symbolizing the completion of a system. A capstone project represents the development of a real-world application that solves practical problems. In this project, the Digital GroceryHub System is designed to simplify grocery shopping through a digital platform. It involves managing products, handling customer orders, and maintaining system activities efficiently. The system provides an easy interface for users to browse items and place orders. Admin manages and controls the overall system operations. Customers can conveniently shop and complete their purchases online. Thus, the project acts as a bridge between theoretical concepts and practical implementation, making the system useful in real-life situations.

Chapter 2

Literature Survey and Specification

2.1 Study of existing system / Review of research paper:

To build an effective Digital GroceryHub System, it is important to understand the current grocery shopping methods and existing systems. Many traditional grocery stores still rely on manual processes where customers physically visit shops to purchase items, which leads to time consumption and inconvenience. In such systems, inventory and order management is often handled manually, resulting in poor data organization, lack of proper tracking, and chances of human errors. Customers also face issues like unavailability of product information, no proper order tracking, and limited digital interaction. These limitations highlight the need for a simple, user-friendly, and efficient system like the Digital GroceryHub System that can manage products, orders, and user activities effectively while improving the overall shopping experience.

We have studied how current grocery systems work. Here's what we found:

Project Co-Ordinator:

Admin:

- Admin can login into the system using secure credentials to access the admin dashboard and manage overall system operations.
- Admin can add, update and delete product details including name, price, description and stock availability.
- Admin can manage customer accounts by viewing user details and handling user-related activities.
- Admin can view all customer orders, update order status and process orders efficiently.
- Admin can manage inventory by tracking stock levels and updating product availability.

Customer:

- Customer can register and login into the system using valid credentials to access all features of the application.
- Customer can browse different categories of products, view detailed information such as price, description and availability.
- Customer can add products to cart, update quantity and remove items before placing the order.
- Customer can place order by providing delivery details and selecting suitable payment method.
- Customer can make online payment and receive confirmation of successful order placement.

Study of exiting project management application

Table 1 Study of existing project grocery applications

Sr No.	Application	Description
1.	BigBasket	Provides online grocery shopping, product categories, delivery tracking, and payment system
2.	Blinkit	Provides fast delivery service, real-time order tracking, and easy product browsing

2.2 Limitations of existing system / Problems discussed in research papers:

i. Dependence on manual and traditional shopping methods
ii. Time-consuming process for customers
iii. Lack of proper inventory and order management
iv. Limited Tracking and transparency in order

2.3 Problem Identification / Need of a system :

i. Customers face difficulty in managing grocery shopping efficiently
ii. Time is wasted in visiting stores and waiting in queues
iii. Vendors face challenges in managing inventory and sales records
iv. Lack of proper system for order tracking and management
v. Difficulty in maintaining customer data and transaction records
vi. Mentor doesn't have right way to evaluate every member
vii. Need for a user-friendly and efficient digital system

2.4 Problem definition:

The existing grocery shopping system lacks automation, real-time inventory tracking, and an integrated online ordering platform. Customers often need to visit stores physically to check product availability, which leads to inconvenience and time loss. Additionally, manual billing and stock management result in errors and stock inconsistencies. Therefore, there is a need to develop a web-based system that allows customers to place grocery orders online while providing real-time stock visibility and automated inventory management for shop owners.

2.5 Specification:

- **Hardware Specification:**
  - Operating System: Windows 11 (64 bit) 16.0GB
  - Processer: Intel(R) Core (TM) i5-3220 CPU @3.30 GHz 3.30 GHz
  - Ram: 16.0GB
- **Software specification:**
  - Server: XAMPP 8.0.28
  - Database: SQLITE

Chapter 3

Scope of Project

The project scope involves developing a comprehensive Digital GroceryHub System that caters to different user roles such as Admin and Customer. The Admin is responsible for managing products, categories, users, and monitoring overall system activities. Customers can browse products, add items to the cart, place orders, track their orders, and manage their purchase history. Non-functional requirements include secure user authentication, real-time order tracking, role-based access control, a user-friendly interface, and efficient system performance. The system aims to simplify the grocery shopping process while ensuring data security, ease of use, and smooth interaction between users.

User roles: Admin, Customer

Functionality:

- Managing product categories
- Adding and updating products
- Managing inventory and stock
- Placing and processing orders
- Tracking order status
- Managing users and roles
- Secure login and authentication
- User-friendly interface
- Real-time updates
- Efficient data management
- Streamlining grocery shopping process
- Easy access and convenience for users

Chapter 4

Methodology

4.1 User Requirement:

Functional Requirement:

There are three user roles for the system: Admin, Client, and Customer.

i. Requirement of Admin:

- Admin able to manage product categories and product list.
- Admin can add, update, and delete products.
- Admin able to manage users (clients and customers).
- Admin can monitor all orders and transactions.
- Admin able to view overall system activity and reports.
- Admin shall be able to maintain records of products and orders.

ii. Requirements of Customer:

- Customer shall be able to login into system with username and password.
- Customer should be able to view available grocery products.
- Customer shall be able to add products to cart.
- Customer shall be able to place orders online.
- Customer shall be able to track order status.
- Customer shall be able to view previous orders.
- Customer shall be able to manage profile and details.

iii. Requirements of User:

- The user shall be able to view the list of available grocery products.
- The system shall display the price of each product clearly.
- The system shall provide information about manufacturer of each product.
- The system shall maintain and display stock history of products.
- The system shall show updated product prices dynamically.
- The user shall be able to add products to the cart using an “Add to Cart” button.
- The user shall be able to view the list of selected items in the cart.
- The user shall be able to update the quantity of selected items.
- The system shall calculate and display the total amount of selected items.

Non Functional Requirement:

- Implement user authentication and authorization for security.
- System should provide real-time order tracking.
- System should support role-based access control.
- System should be user-friendly and easy to use.
- System should be efficient and fast in processing data.
- System should maintain proper data records for future use.

4.2 Block Diagram of System:

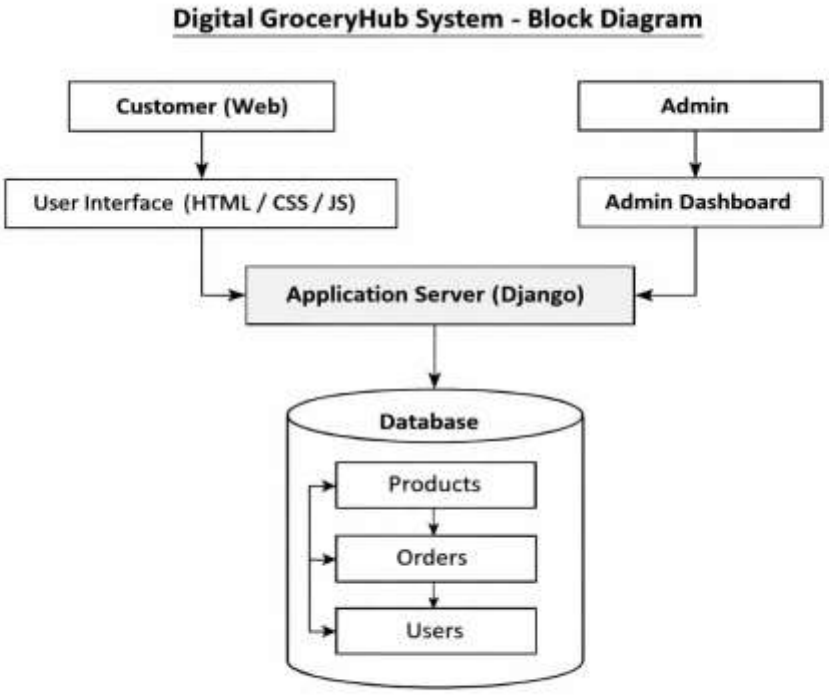

Figure 4.1 Block diagram of Digital GroceryHub System

Figure 4.1 shows the proposed system diagram of Digital GroceryHub System, where the diagram is divided into three parts: input required for the system, output provided by the system, and database used to store grocery-related data.

Input: Customer details consist of all relevant data such as name, email, mobile number, address, and password. Admin details include admin username and password for system access. Product details contain product name, category, price, quantity, and description. Order details consist of order ID, product list, quantity, total amount, payment status, and delivery address.

Output : The system will provide product listings with details for customers. It will generate order confirmation with order ID and details. The system will display order status such as pending, processing, and delivered. It will also provide order history for customers and reports for admin. Additionally, the system offers easy navigation and quick access to product and order information for better user experience.

4.3 Action Plan:

Steps followed to developed Digital GroceryHub System :

i. Finalization of project team and allocation of guide
ii. Identify project domain/area (online grocery system)
iii. Submission of project proposal/project ideas by the project team
iv. Finalization of project idea by project guide and HOD
v. Industrial survey and Literature survey
vi. Project Identification and Specification
vii. Proposed detailed methodology of solving the identified problem
viii. with action plan
ix. Preparation of project planning report and presentation
x. Presentation
xi. Submission of project planning report
xii. Design User Interface
xiii. Design database structure, create database
xiv. Develop GUI and various Interfaces
xv. Programming / coding
xvi. Database connectivity
xvii. Installation of Project, Testing
xviii. Modifications as per suggestions given by guide and course teacher
xix. Prepare project presentation, Compile a draft copy of complete
xx. Project Report
xxi. Make correction in draft copy of project report, Presentation and
xxii. Submission of project report and term work

4.4. Technology use to develop project:

- Structure: HTML
- Design: CSS
- Validation: JAVASCRIPT
- Database: SQLITE
- Backend: Django python

Chapter 5

Details of designs, working and processes

5.1 System Flow Diagram:

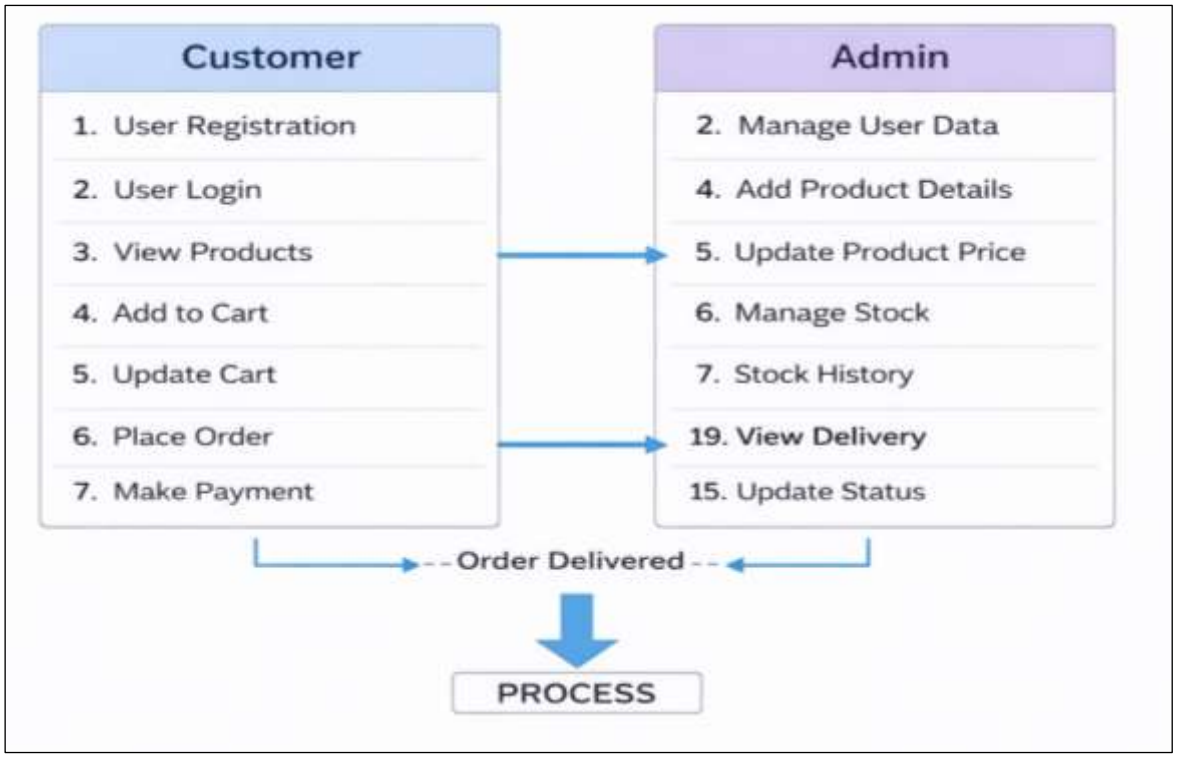

Figure 5.1 Flow diagram of Digital GroceryHub System

Figure 5.1, shows the flow diagram of the Digital GroceryHub System with two user roles: Customer and Admin. The system begins with customer registration and login. Customers can view products, add items to the cart, and update the cart. After that, the customer places an order and proceeds with payment. The admin manages user data and adds product details in the system. Admin also updates product prices and manages stock availability. The admin views delivery details and updates the order status accordingly. Finally, the order is completed successfully after coordination between the customer and admin.

5.2 Site Map:

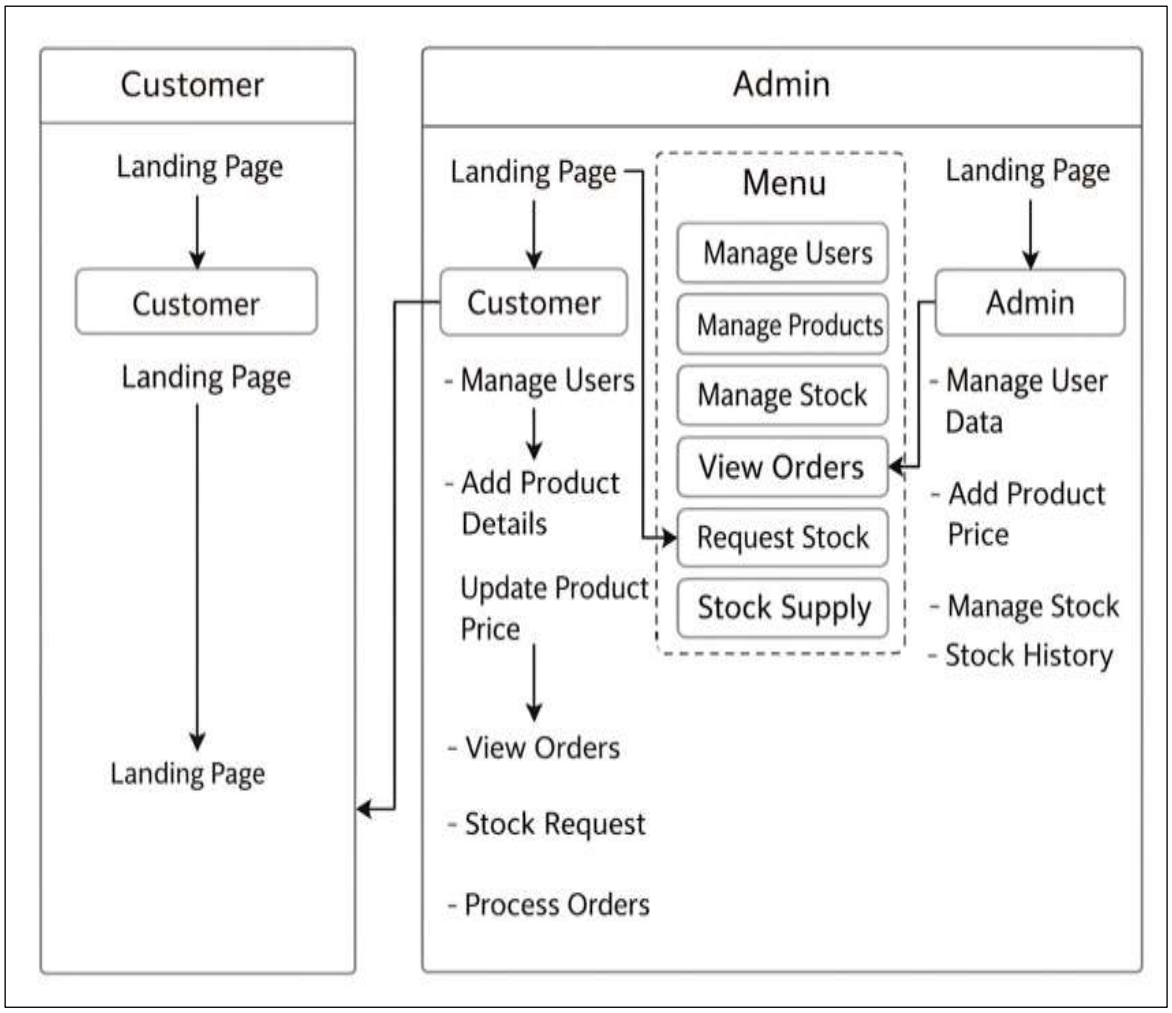

Figure 5.2 Site Map of Digital GroceryHub System

Figure 5.2 illustrates the site map and navigation flow of the Digital GroceryHub System, where the Landing Page serves as the common entry point for all users. Based on their roles as Customer or Admin, users are directed to their respective sections and gain access to role-based functionalities. The Admin dashboard provides a centralized menu to manage users, products, and stock, view and process orders, and handle stock requests. Meanwhile, the Customer interacts with the system through the landing page to access services and manage orders. This role-based structure ensures a secure, organized workflow and simplifies navigation, thereby enhancing the overall user experience.

6 Admin Module Flow:

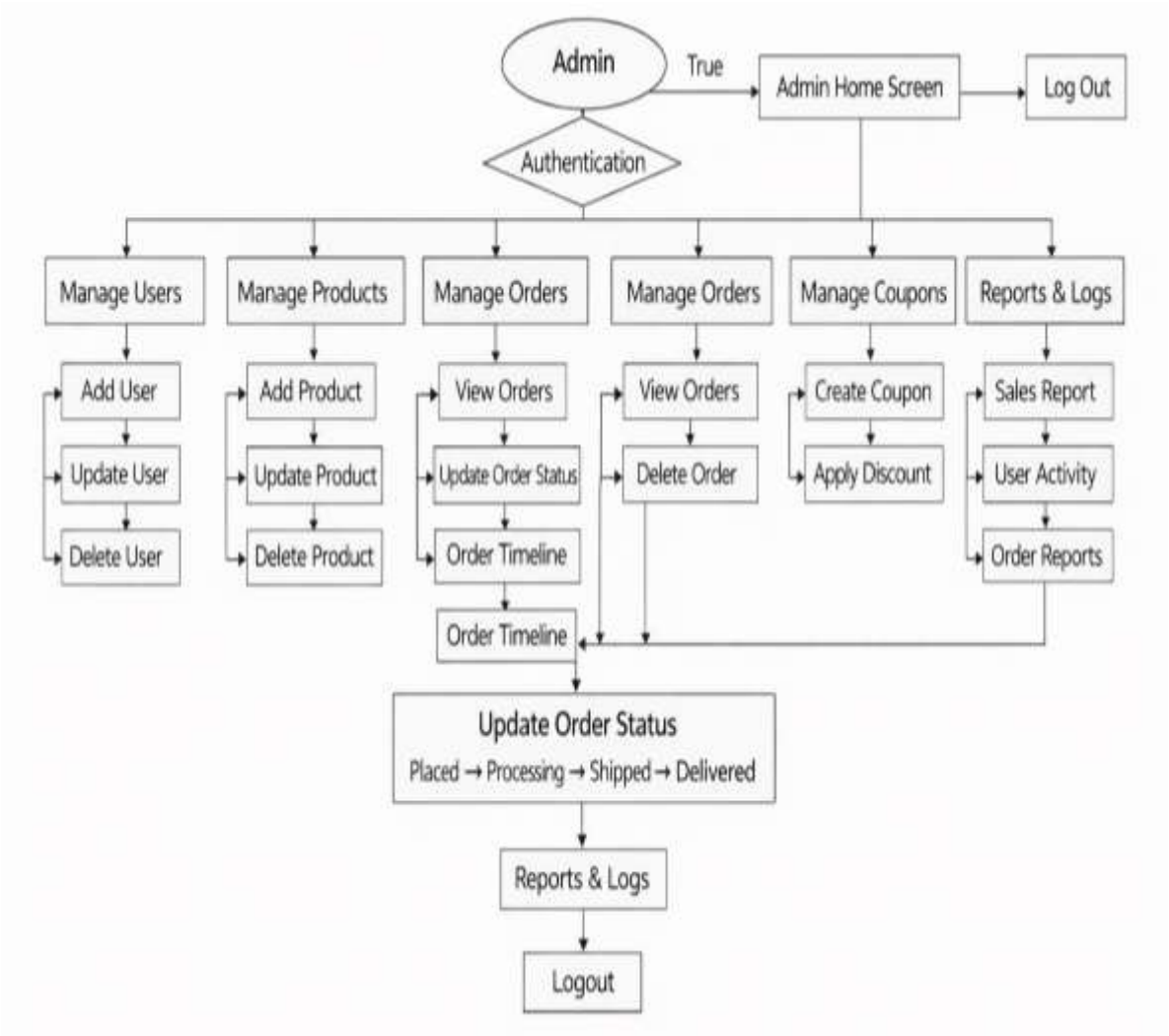

Figure 5.3 Admin module flow

Figure 5.3, shows the Admin module flow where from the menu the admin is redirected to manage various functionalities such as users, products, orders, coupons, and reports. In user management, the admin can view, add, update, and delete user details. In product management, the admin can add new products, update product information, delete products, and manage stock availability. In order management, the admin can view orders, update order status, delete orders, track the order timeline. In coupon management, the admin can create coupons, apply discounts, and manage their usage. In reports and logs, the admin can generate sales reports, monitor user activity, and view order reports. Additionally, through order timeline management, the admin can track and update order status across different stages such as placed, processing, shipped, and delivered.

6.1 Customer Module Flow:

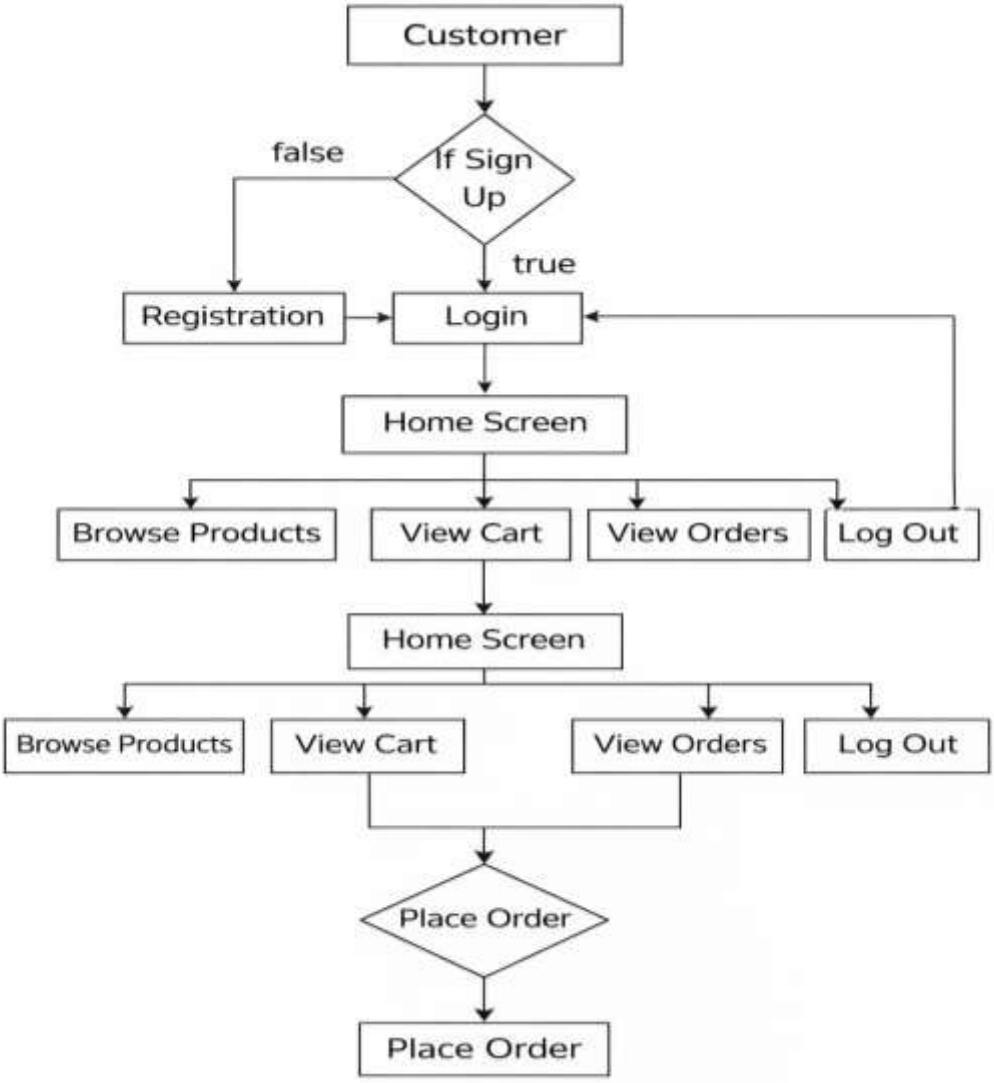

Figure 5.4 Customer module flow

Figure 5.4, shows the flow of customer activities starting from registration and login. After successful login, the customer can browse products and view product details. The customer can add items to the cart and proceed to place an order. Finally, the system allows the customer to make payment and track order status, showing the complete interaction flow within the module.

6.2 E-R Diagram:

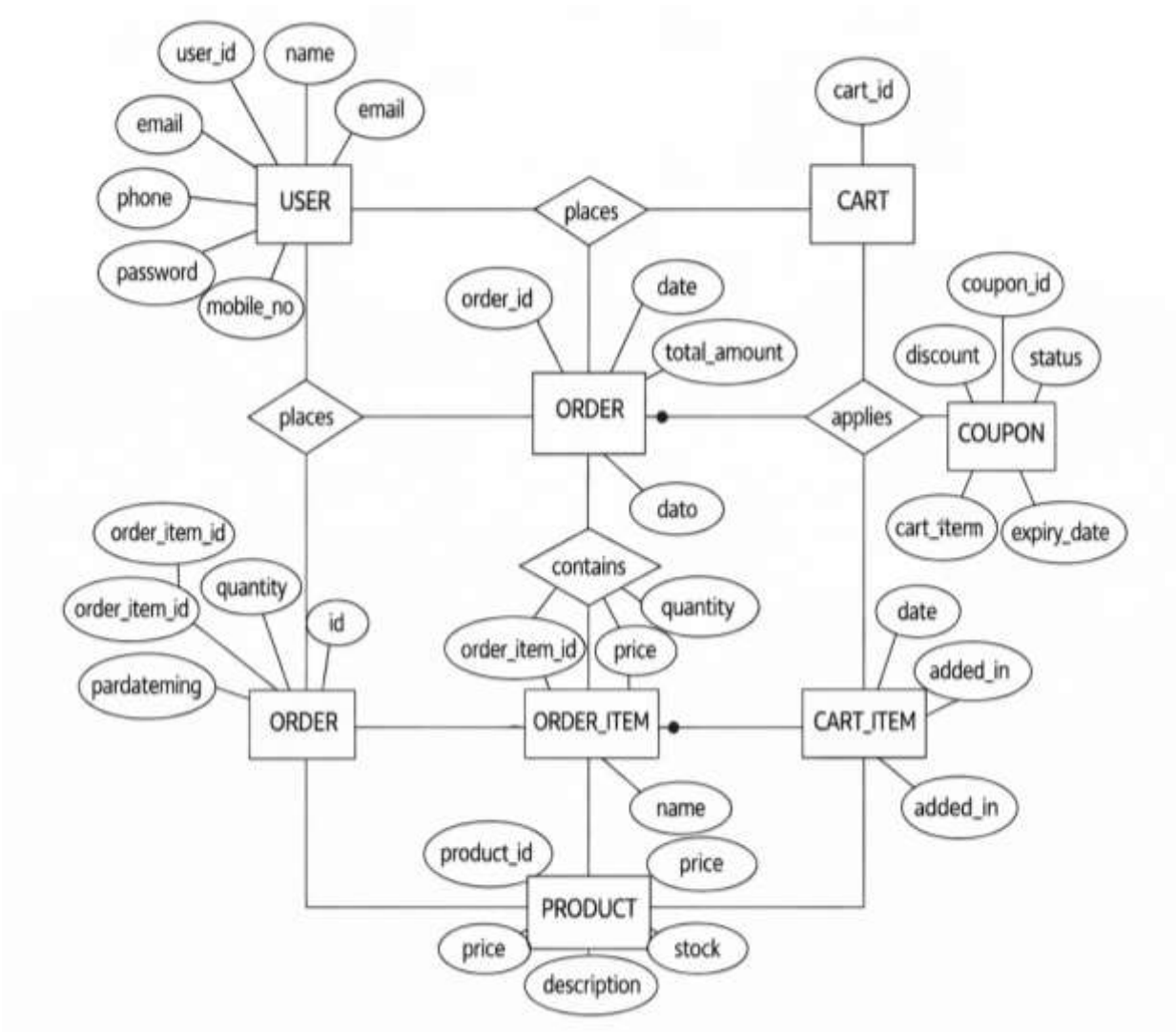

Figure 5.5 E-R diagram of Digital GroceryHub System

Figure 5.5, Shows the represents the structure of the Digital GroceryHub system. It illustrates the relationship between various entities such as User, Cart, Order, Product, Coupon, Cart_Item, and Order_Item.A user can place orders and manage a shopping cart. The cart contains multiple cart items, and each cart item is associated with a specific product. Similarly, an order consists of multiple order items, where each item represents a product purchased by the user.The system also allows applying coupons to the cart for discounts. Each entity is connected through meaningful relationships like “places,” “contains,” and “applies,” ensuring smooth data flow and proper organization of the system.This ER diagram helps in understanding the database design and how different components of the system interact with each other, making it easier to implement and manage the application efficiently.

Table 2 Database table description

Sr No.	Table Name	Fields Name	Purpose
1	Customer	Id, firstname, lastname, email, mobileno, address, password, status	Customer table contains customer personal information used for login and order processing.
2	Admin	Id, username, password	Admin table contains admin login credentials to manage system.
3	Category	Id, categoryname, description	Category table is used to organize products into different groups.
4	Product	Id, productname, category, price, quantity, description, image	Product table contains all grocery product details available in system.
5	Orders	Id, orderid, productid, quantity, price	Orders table contains order summary placed by customer.
6	OrderDetails	Id, orderid, productid, quantity, price	OrderDetails table contains detailed information of each product in an order.

Sr No.	Table Name	Fields Name	Purpose
7	Payment	Id, orderid, paymentmethod, paymentstatus, transactionid	Payment table contains payment-related details of orders.
8	Delivery	Id, orderid, deliveryaddress, deliverydate, status	Stores delivery details

Functionality provided by System:

1. User Management
2. Product Management
3. Order Management
4. Delivery Management
5. Progress Tracking

1. User Management:

The User Management module handles information about users such as Admin, Client, and Customer. Each user has specific access based on their role. Admin manages all users, clients manage products and orders, and customers can browse and purchase products. This module helps in organizing user data and maintaining system security.

Step wise procedure of user management module:

Step 1. Registration of client and customer through the form

This step allows users to create an account by entering required details. It stores user information in the system database. It is the first step to access the system.

- Customer needs to fill details: Name, email, mobile number, address, password

Step 2. Admin can view all customer and client records on dashboard

Admin can see all registered users in one place. It helps in monitoring user data.

Step 3. Admin can apply filters to sort records in ascending or descending order

Filtering helps to manage large data easily. It improves searching and sorting of records. It saves time for admin.

Step 4. **Admin can update or delete user details**

Admin can correct or remove user data. It ensures data accuracy. It maintains system integrity.

Step 5. Same process is applicable for both client and customer records

System follows same steps for all users. It keeps process simple and uniform.

Implementation of User Management Module:

Algorithm:

Step 1: Start

Step 2: Create customer table in database

Step 3: Create customer table in database

Step 4: Add customer record through registration form

Step 5: Add customer record through registration form

Step 6: Display all records in table format

Step 7: Add update and delete option for each record

Step 8: If update clicked, update record

Step 9: If delete clicked, delete record

Step 10: End

2. Product Management:

The Product Management module is responsible for handling all activities related to products in the Digital GroceryHub System. It allows admin or client to add, update, delete, and manage product details efficiently. This module ensures that all grocery items are properly organized and categorized for easy access. It helps in maintaining accurate product information such as price, quantity, and description. The system stores product data securely in the database for future use. It also ensures that only available products are displayed to customers. This improves the overall shopping experience for users. The module provides flexibility to update product details whenever required. It reduces manual work and improves system efficiency.

Step wise procedure of Product Management module:

Step 1. Admin adds product details (name, category, price, quantity)

This step allows admin or client to enter product information into the system. It helps in maintaining product records. It is the first step to make products available online.

Step 2. Product is stored in database

All product details are saved securely in the database. It ensures data availability for future use. It helps in managing product information efficiently.

Step 3. Products are displayed to customers

Stored products are shown on the user interface. Customers can easily browse available items. It improves user experience and accessibility.

Step 4. Admin can update or delete product

Admin or client can modify or remove product details. It helps to keep product data updated. It ensures accuracy of information.

Algorithm for product management:

Step 1: Start

Step 2: Enter product details such as name, category, price, quantity

Step 3: Store product details in product table

Step 4: Display product list in table format

Step 5: Provide option to update or delete product

Step 6: End

Algorithm for order processing:

Step 1: Start

Step 2: Select products from product list

Step 3: Add selected products to cart

Step 4: Confirm order and store details in order table

Step 5: Update order status (pending, processing, delivered)

Step 6: End

Algorithm for delivery management:

Step 1: Start

Step 2: Select order from order table

Step 3: Assign delivery details

Step 4: Update delivery status

Step 5: Display delivery status to customer

Step 6: End

3. Order Management:

The Order Management module acts as a central system for handling customer orders. It allows customers to select products, place orders, and track their status. Admin or client can view, process, update, and manage all orders efficiently. Each order includes details such as products, quantity, total amount, and status. This module helps in smooth processing and tracking of orders from placement to delivery.

Stepwise procedure of Order Management:

- *From Admin/Client perspective:*

Step1: Select order management option

Step 2: View list of all customer orders

Step 3: Check order details such as products and quantity

Step 4: Update order status (pending, processing, delivered)

Step 5: If any issue, update or modify order

Step 6: Repeat process until order is completed

Step 7: Admin can view all orders history

Step 8: End

- *From Customer perspective:*

Step1: Browse products

Step 2: Add products to cart

Step 3: Place order

Step 4: View order details

Step 5: Track order status

Step 6: Receive updates from system

Step 7: View previous orders

Step 8: End

Implementation of Order Management:

Step 1: Start – Initializes the process.

Step 2: Add order option in system – Provides interface to place order.

Step 3: Store data in order table – Saves order details in database.

Step 4: Store product details – Maintains product-wise order data.

Step 5: Display order details – Allows admin/client to view orders.

Step 6: Update order status option – Enables tracking of order.

Step 7: Update status in database – Saves latest order status.

Step 8: Display status to customer – Shows current order status.

Step 9: Maintain order history – Stores previous order records.

Step 10: End – Completes the process.

4. Delivery Management :

Delivery Management in Digital GroceryHub System ensures smooth and timely delivery of orders to customers. This module helps in assigning delivery tasks, tracking delivery status, and managing delivery agents. It ensures that orders are delivered safely and efficiently while keeping customers updated about their order status.

Stepwise procedure of Delivery Management:

Step 1: Order is ready for delivery after packing
Step 2: System assigns delivery to delivery agent/client
Step 3: Delivery details are displayed to assigned person
Step 4: Delivery agent picks up the order
Step 5: Order is out for delivery
Step 6: Customer receives delivery notification
Step 7: Delivery agent reaches customer location
Step 8: Order is delivered successfully
Step 9: Delivery status is updated in system
Step 10: End

Implementation of Delivery Management:

Algorithm for Delivery Assignment:

Step 1: Start – Initialize delivery assignment

Step 2: Fetch all orders ready for delivery

Step 3: Check list of available delivery agents

Step 4: Select suitable agent based on availability/location

Step 5: Assign order to selected agent

Step 6: Update assignment details in database

Step 7: Send notification to delivery agent

Step 8: Wait for acceptance from agent

Step 9: If rejected, assign to another agent

Step 10: End

Algorithm for Delivery Tracking

Step 1: Start – Begin tracking process

Step 2: Select order for tracking

Step 3: Fetch delivery details from database

Step 4: Display current delivery status

Step 5: Update status as “Picked”, “Out for Delivery”

Step 6: Track location of delivery agent (if available)

Step 7: Notify customer with updates (SMS/App)

Step 8: If delivery failed, mark as “Failed Attempt”

Step 9: Reschedule delivery if required

Step 10: Once delivered, update status as “Delivered”

Step 11: Store final delivery record

Step 12: Display final status to user

Step 13: End

5. Progress Tracking:

The main aim of progress tracking in Digital GroceryHub System is to visualize the status of orders and deliveries. The system uses color codes to represent different stages of order processing. This helps admin, client, and customer to quickly understand order status such as completed, pending, or in progress.

Implementation of progress tracking of order status:

Step 1: Start

Step 2: Display blocks for each order stage – Each block represents a stage like placed, processing, shipped, delivered.

Step 3: Fetch order status from database – Retrieves current order status.

Step 4: Apply color logic for each stage:

1. If order is delivered → show green (completed)
2. If order is pending → show white/blank
3. If order is in processing → show blue (current stage)
4. If order is delayed → show red (not completed)

Step 4: Repeat for all orders – Ensures tracking for every order

Implementation of progress tracking of every individual work:

Step 1: Start

Step 2: Retrieve order data – System fetches order details from order table based on Customer ID.

Step 3: Order table contains fields like order id, customer id, product details, quantity, date, and status.

Step 4: Display order list – All orders of customer are shown in dashboard in structured format.

Step 5: Check status of each order – System reads status field to identify current stage.

Step 6 If no order found – Display blank or “No Orders” message to user.

Step 7: If order exists, apply color coding logic:

1. Status = placed → gray (order successfully placed)
2. Status = processing → blue (order is being prepared)

3. Status = shipped → orange (order is out for delivery)
4. Status = delivered → green (order completed successfully)
5. Status = cancelled → red (order cancelled or failed)

Step 8: Update status dynamically – Status changes automatically as order moves to next stage.

Step 9: Display updated status to customer – Customer can track real-time progress of order.

Step 10: Maintain order history – All previous orders are stored for future reference.

Step 11: Repeat for all orders – Ensures tracking of every customer order.

Step 12: End – Completes the process.

Chapter 6

Result and Application

6.1 Result:

Registration Form Web Page for User :

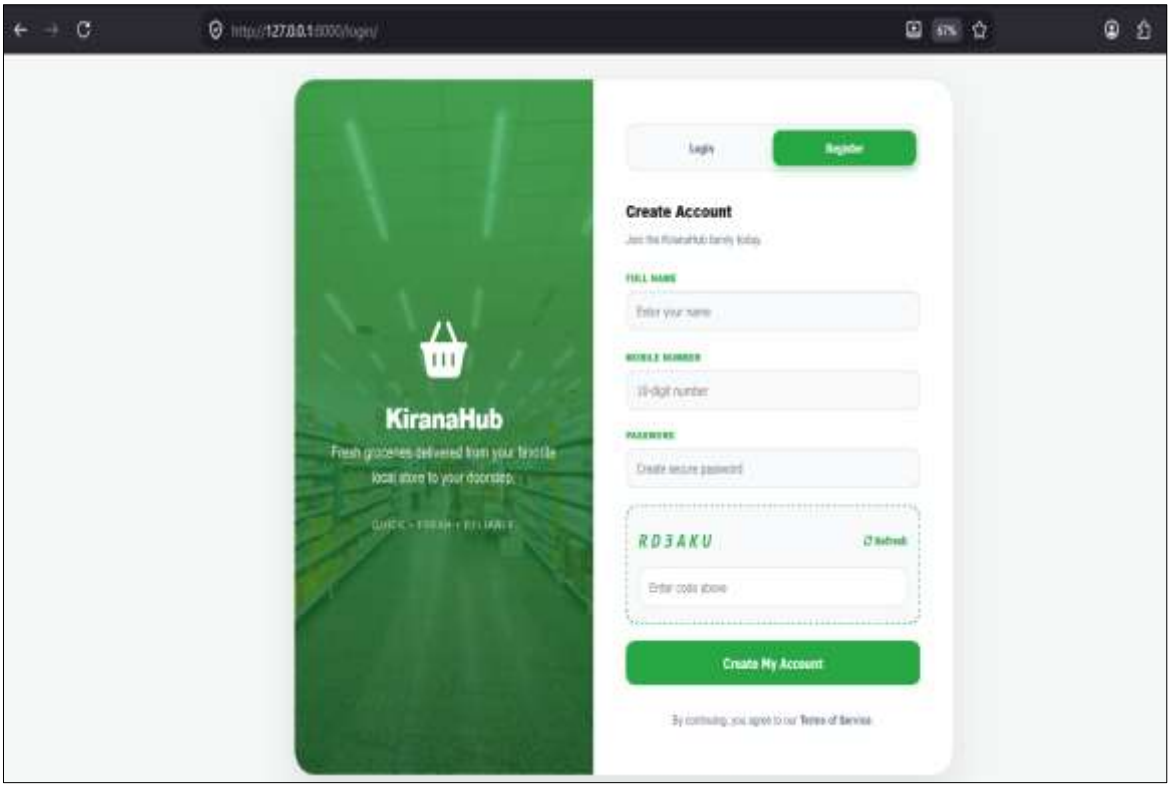

Figure 6.1 Registration form for User

User data is entered through the above registration form in the system. The data is filled by the user itself.

Table 3 User registration form validations

Fields	Validation Applied
Name	"[A-Za-z]+", "Only alphabetic characters allow"
Email ID	Valid email format required (e.g., example@gmail.com)
Mobile Number	"[789]\d{9}", Mobile number must start with 7, 8, or 9 and must be 10 digits
Password	"(?=.\d)(?=.[a-z])(?=.*[A-Z]).{8,16}", Must contain at least one number, one uppercase and one lowercase letter, and be 8–16 characters long
CAPTCHA	Must match the generated CAPTCHA code
Terms & Conditions	User must accept terms and conditions

Login Page:

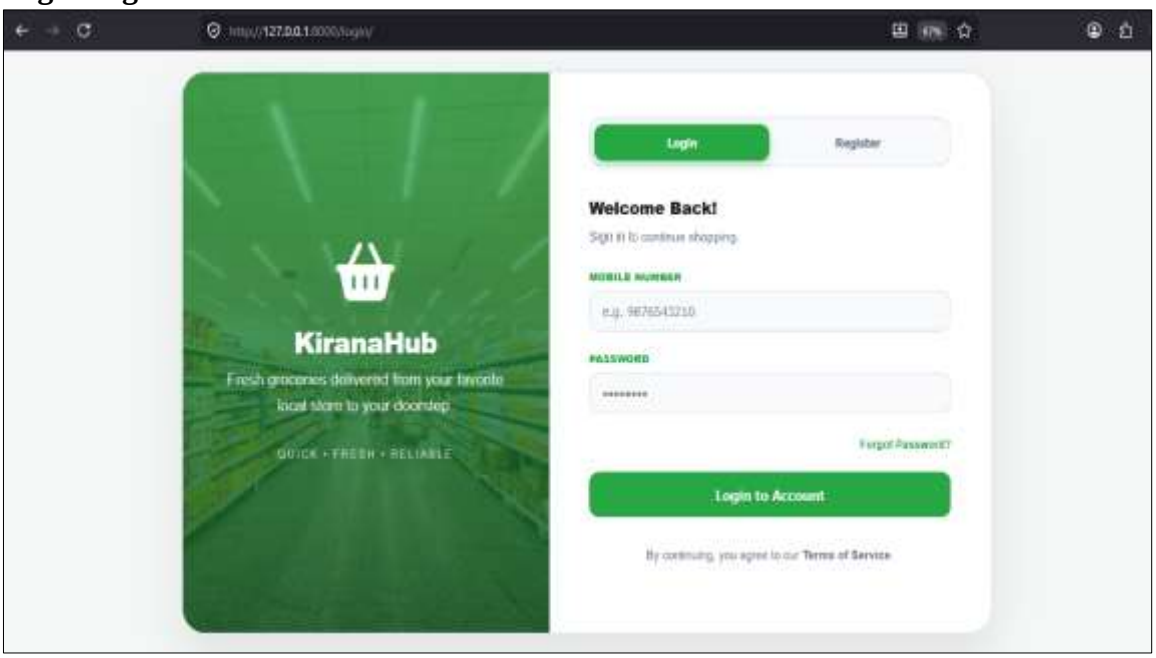

Figure 6.2 Login Page for user

Login page of user accepts mobile number and password to allow only authorized user to access the system dashboard.

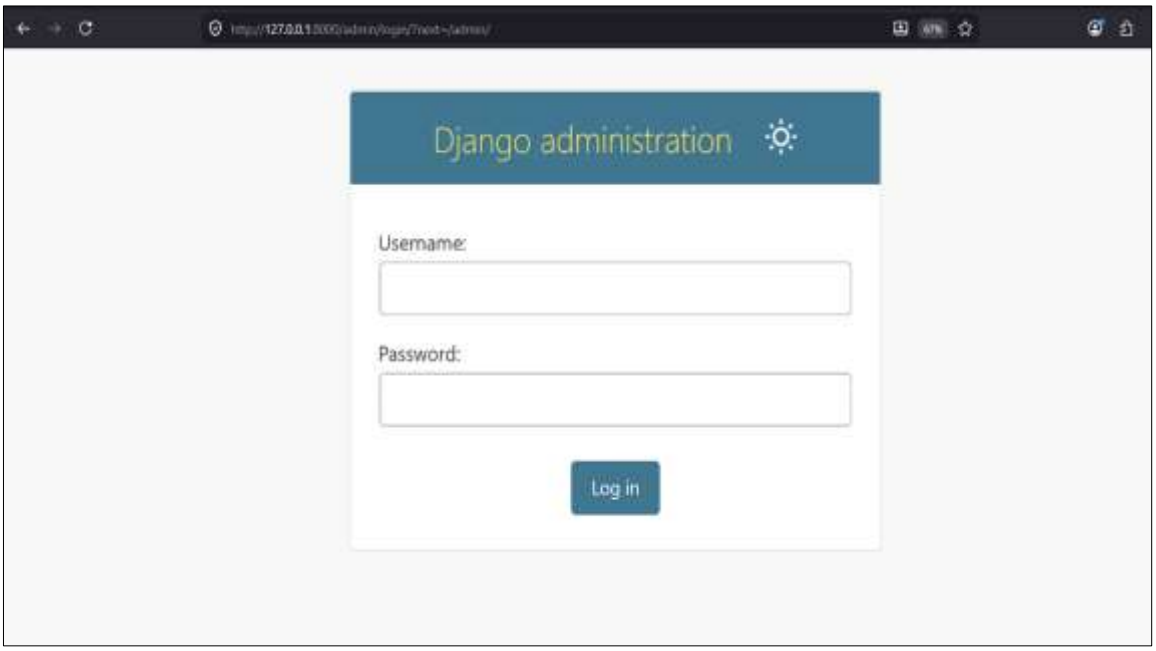

Figure 6.3 Login Page for Admin user

Login page of admin accepts admin mobile number and password to allow only authorized admin to access the admin dashboard and manage system functionalities.

Admin Module:

1. Home Page -Admin:

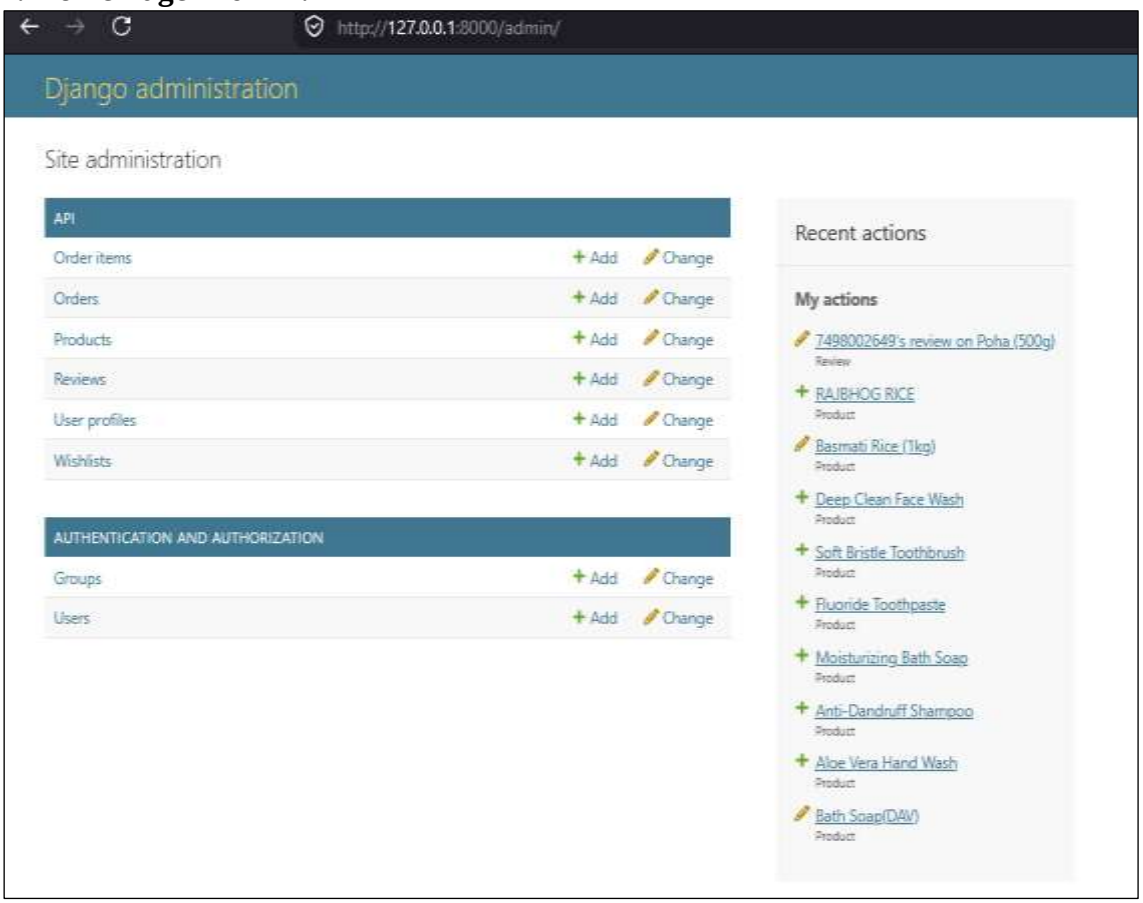

Figure 6.4 Admin Home Page

- Admin home page contain header at top and menu at left side. The page shows Django administration dashboard where admin can manage all system modules. Left side shows modules like Order Items, Orders, Products, Reviews, User Profiles and Wishlists where admin can Add and Change data. Admin can also manage authentication using Groups and Users section. Right side shows recent actions which display latest activities performed by admin.
- Admin can add and update records.
- Admin can manage overall system from this page.

2. Manage Item –Admin:

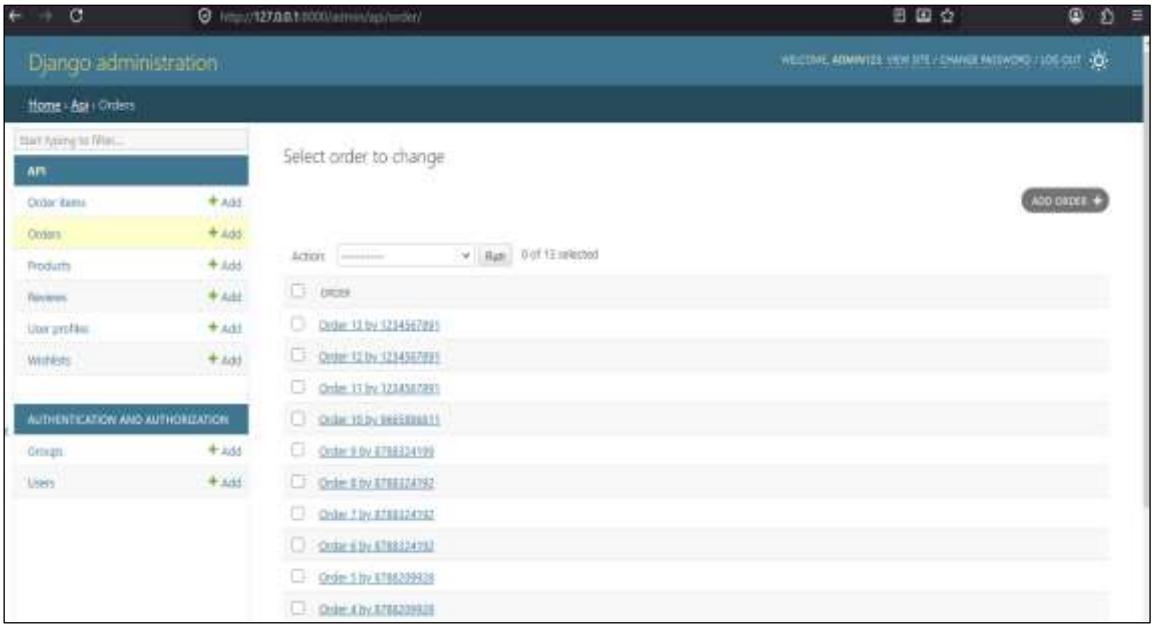

Figure 6.5 Manage Item Page - Admin

- All order item records are displayed in the admin panel.

- Admin can view and manage order item details such as product and quantity.

- Admin can add new order items and perform actions on selected records.

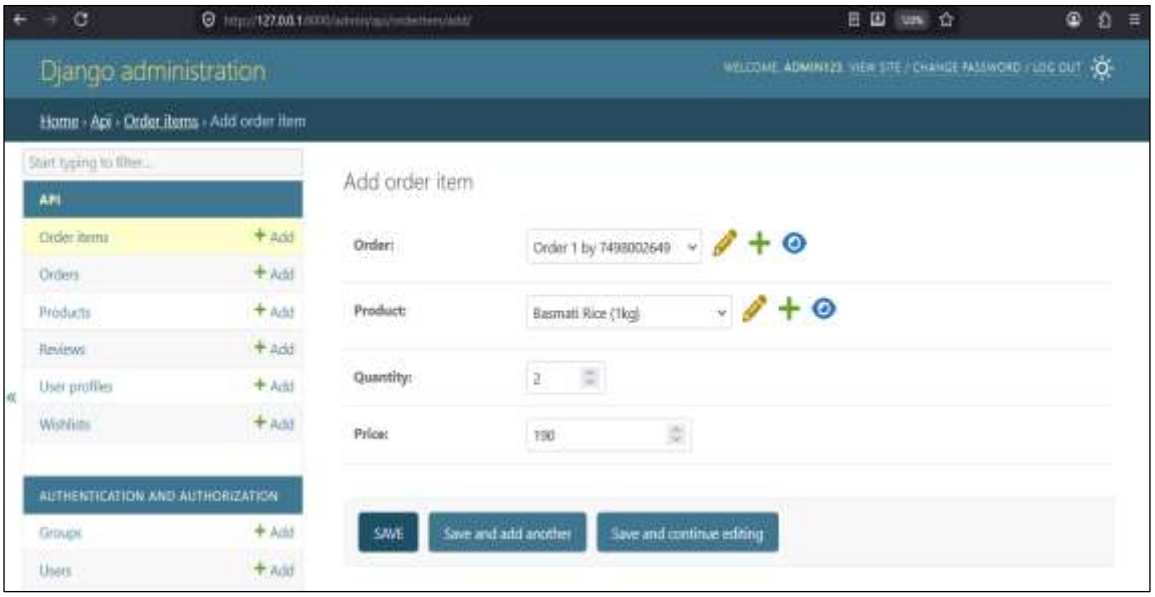

Figure 6.6 Add Order Item – Admin

- All order item details are displayed including order, product, quantity, and price.

- Admin can add, update, and delete order item records easily.

- Dropdown selection and save options make data entry simple and error-free.

3. Order Management – Admin:

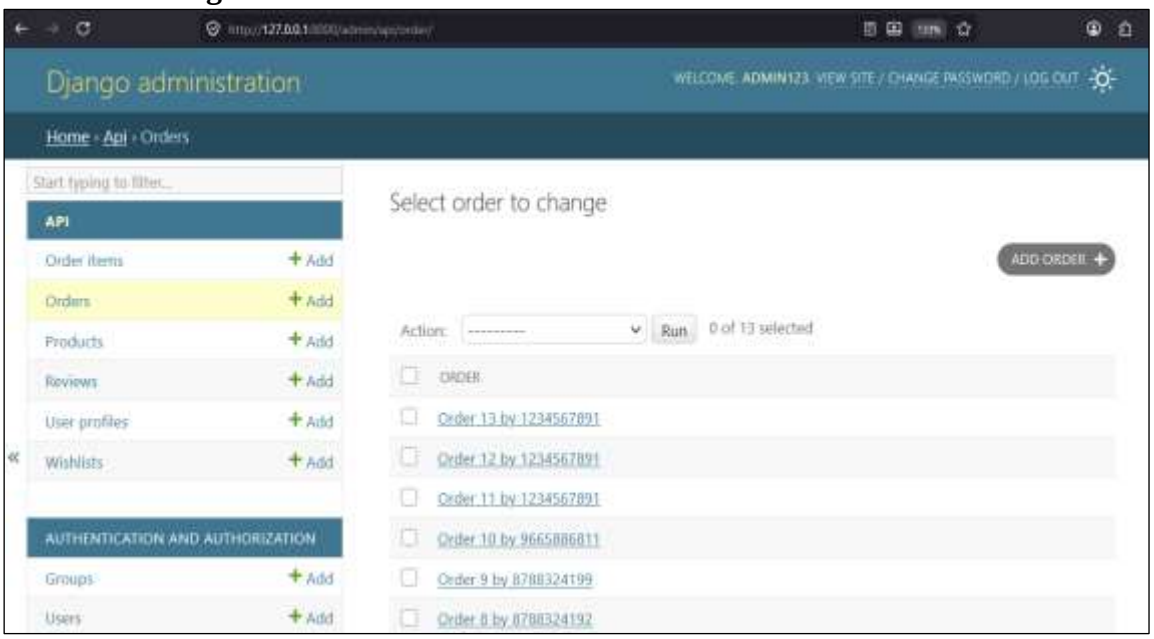

Figure 6.7 Order management – Admin

- All order details are displayed here including user, total amount, discount, and status.

- Admin can add, view, and update order information easily.

4. Product– Admin:

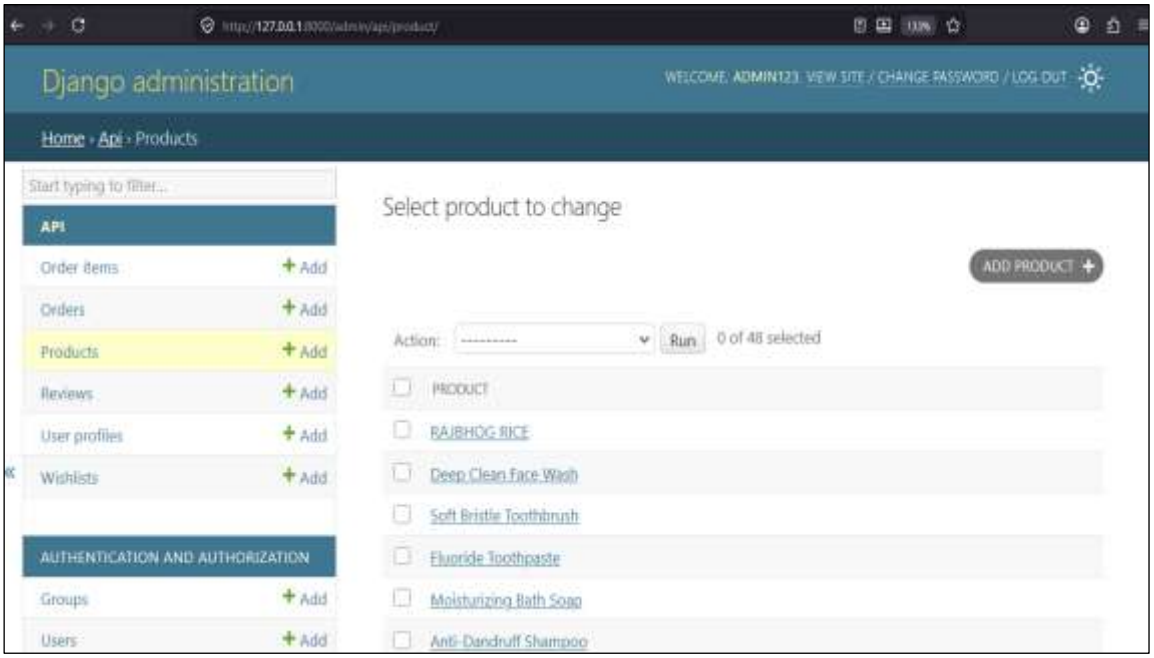

Figure 6.8 Product– Admin

All product details are entered here including name, price, category, and stock, admin can add and update product information easily, and description and image fields help in better product representation.

5. Product Review– Admin:

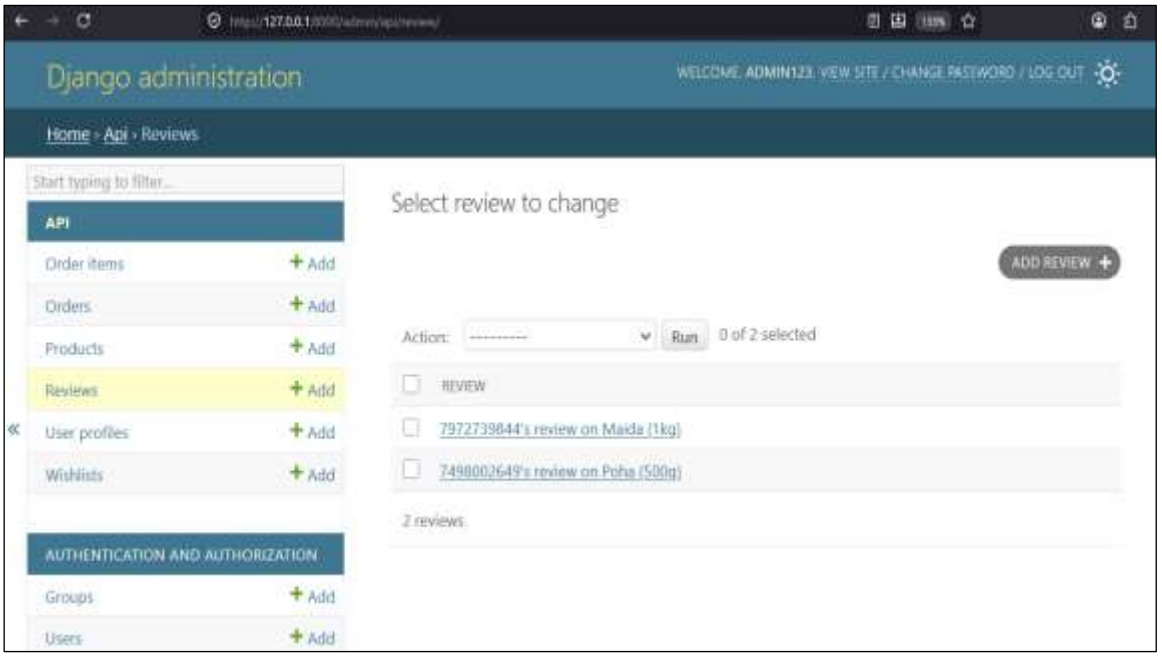

Figure 6.9 Add Product Review – Admin

- Admin can select product and user, and add rating and review comment
- Admin can enter feedback details to manage product reviews easily

6. User – Admin

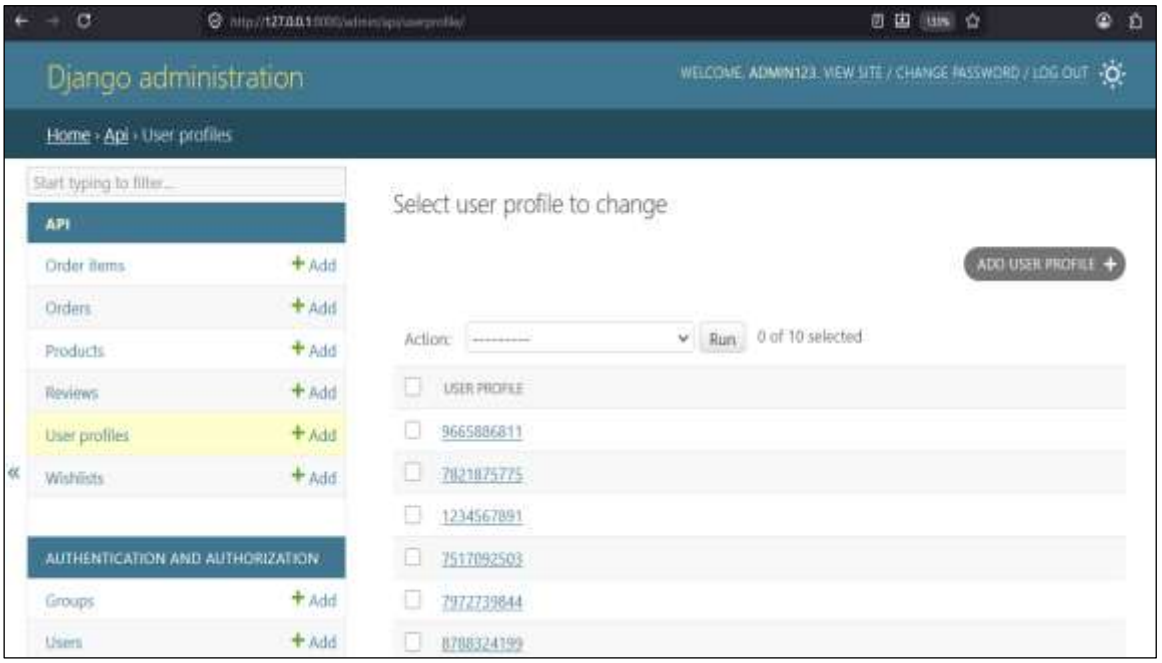

Figure 6.10 User – Admin

7. Wishlists– Admin:

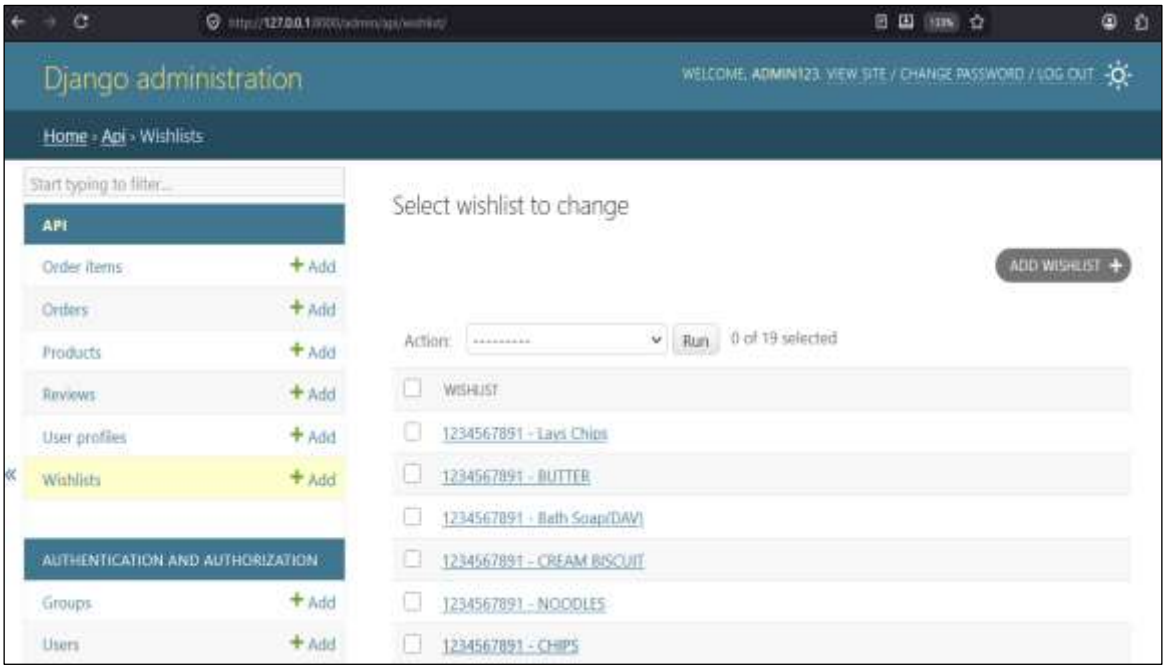

Figure 6.11 Wishlists - Admin

- Admin can select user and product add items to wishlist and manage user preferences easily

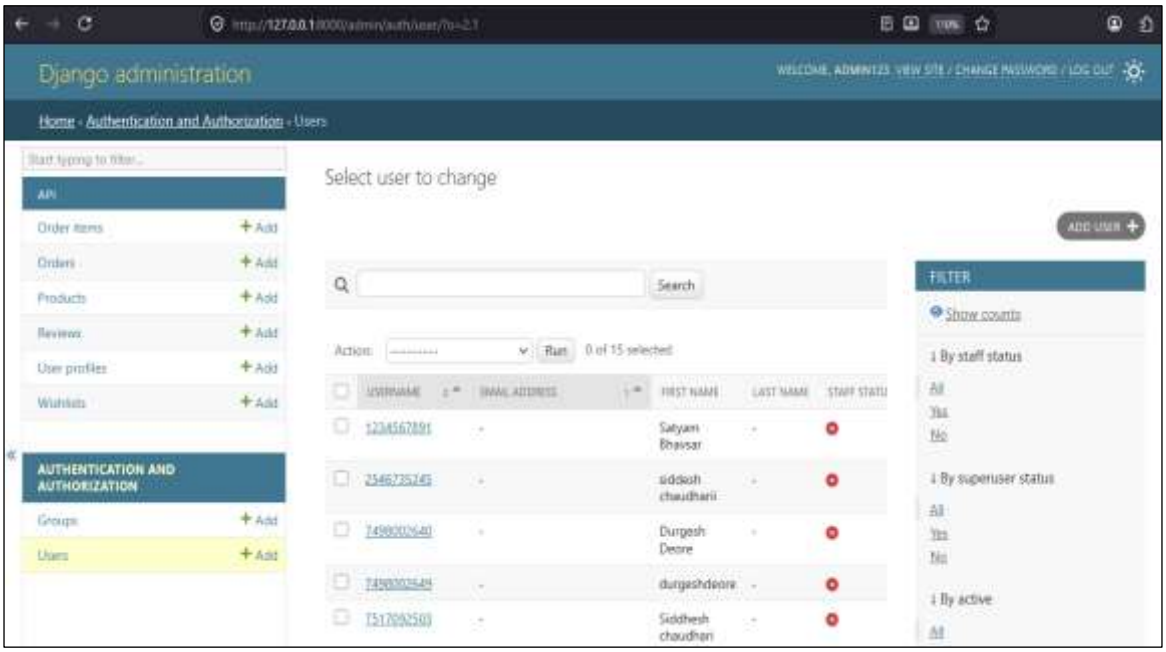

Figure 6.12 User Authentication – Admin

Customer Module:

1. Home Page – Customer:

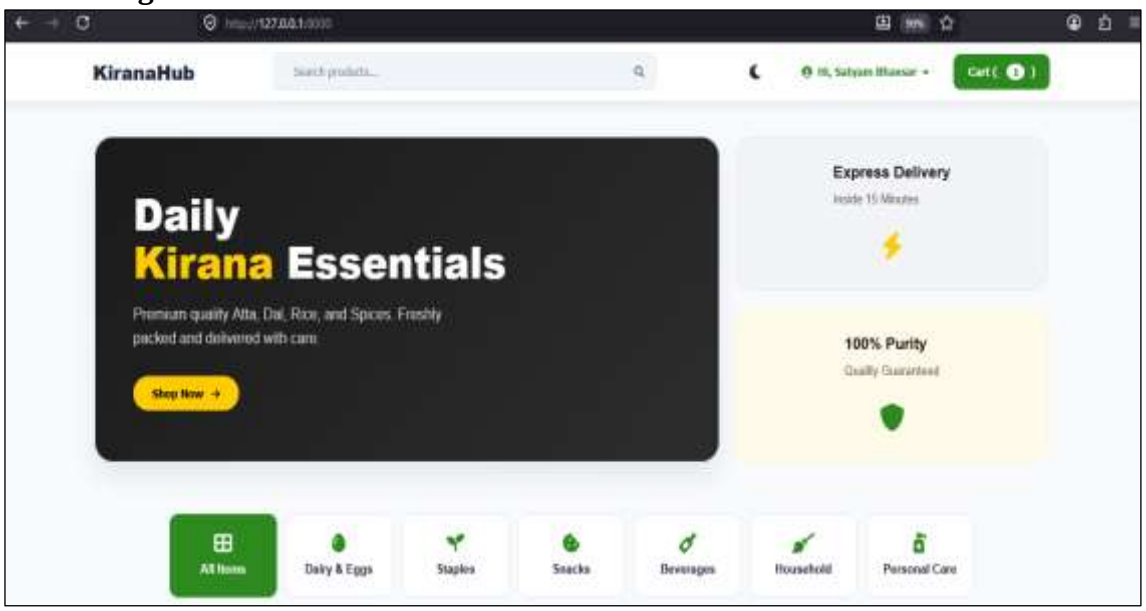

Figure 6.13 Home Page – Customer

- Customer can view product categories.
- Each category/group has two display options to manage and control its content separately for better usability.

2. Featured Products Section– Customer:

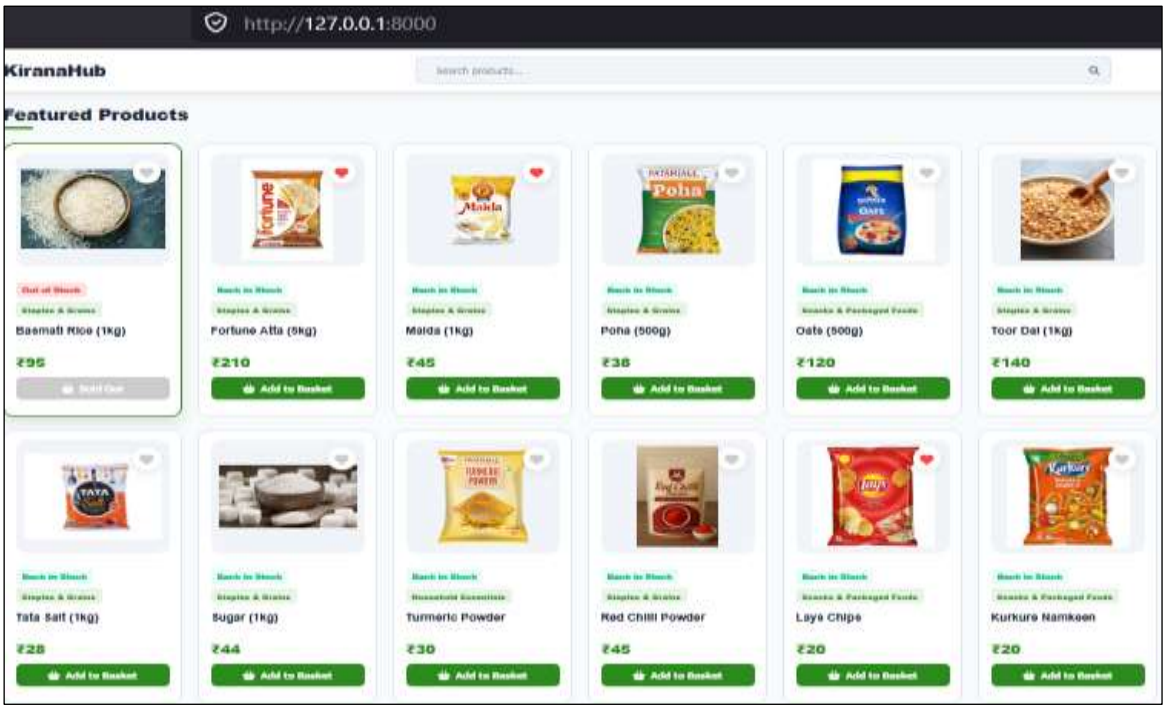

Figure 6.14 Featured Products Section– Customer

Features of Product Management Module:

a. a. Product categories are automatically displayed, and updated dynamically when new products are added or removed.
b. b. Color coding is used to show product availability status as follows:
c. Gray – Out of Stock (Product currently unavailable)
d. Red – Low Stock (Limited quantity remaining)
e. Blue – Newly Added (Recently added product)
f. Green – Available (Product in stock and ready to purchase)
g. c. Each category's products are managed using a status column in the table.
h. d. Redirect user/admin to manage or view details of a specific product.

Code that shows product status is as follows:

```
$selectPQuery = "SELECT * FROM `products` WHERE
category_id=$cid";
$PResult = mysqli_query($con, $selectPQuery);

if (mysqli_num_rows($PResult) > 0) {
   while($Pchecked = mysqli_fetch_assoc($PResult)) {

      if($Pchecked['status'] == 'low'){
         // Red - Low Stock
      }
      elseif($Pchecked['status'] == 'available'){
         // Green - Available
      }
      elseif($Pchecked['status'] == 'new'){
         // Blue - Newly Added
      }
      else{
         // Gray - Out of Stock
      }

   }
}
else{
   // No products found
}
```

3. Product Details View – Customer:

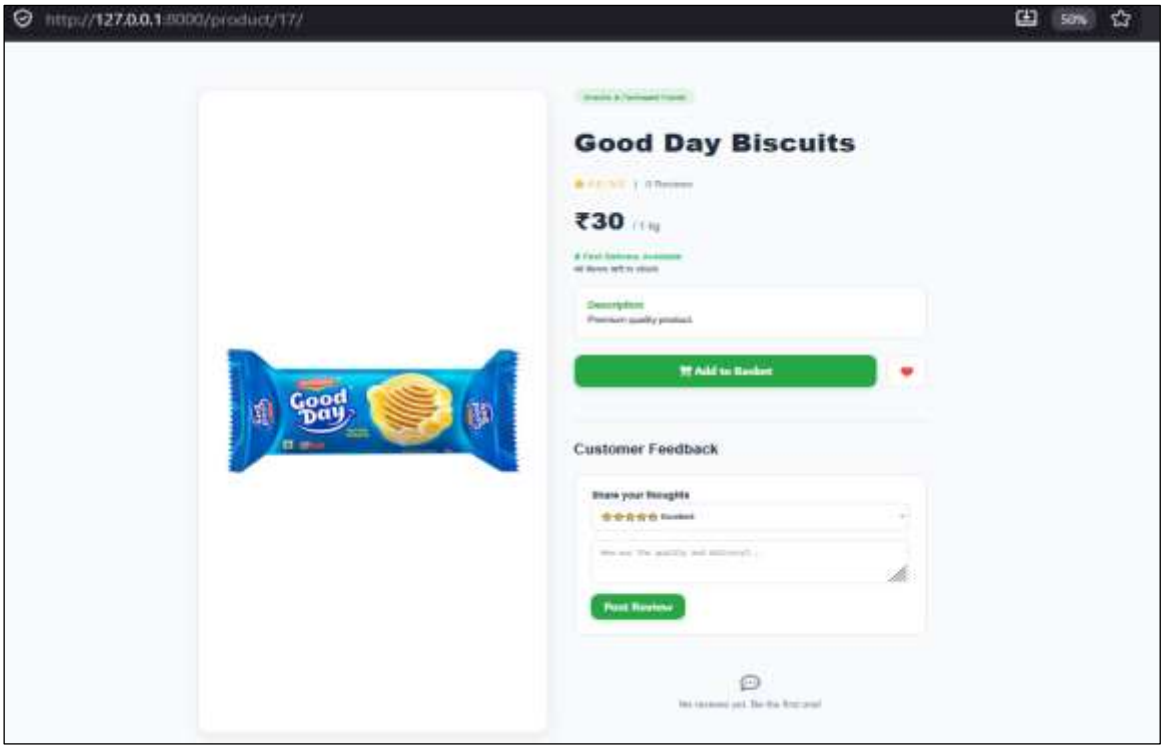

Figure 6.15 Product Details View– Customer

- It includes an “Add to Basket” option and delivery availability details.
- Customers can give feedback and submit reviews in the feedback section.

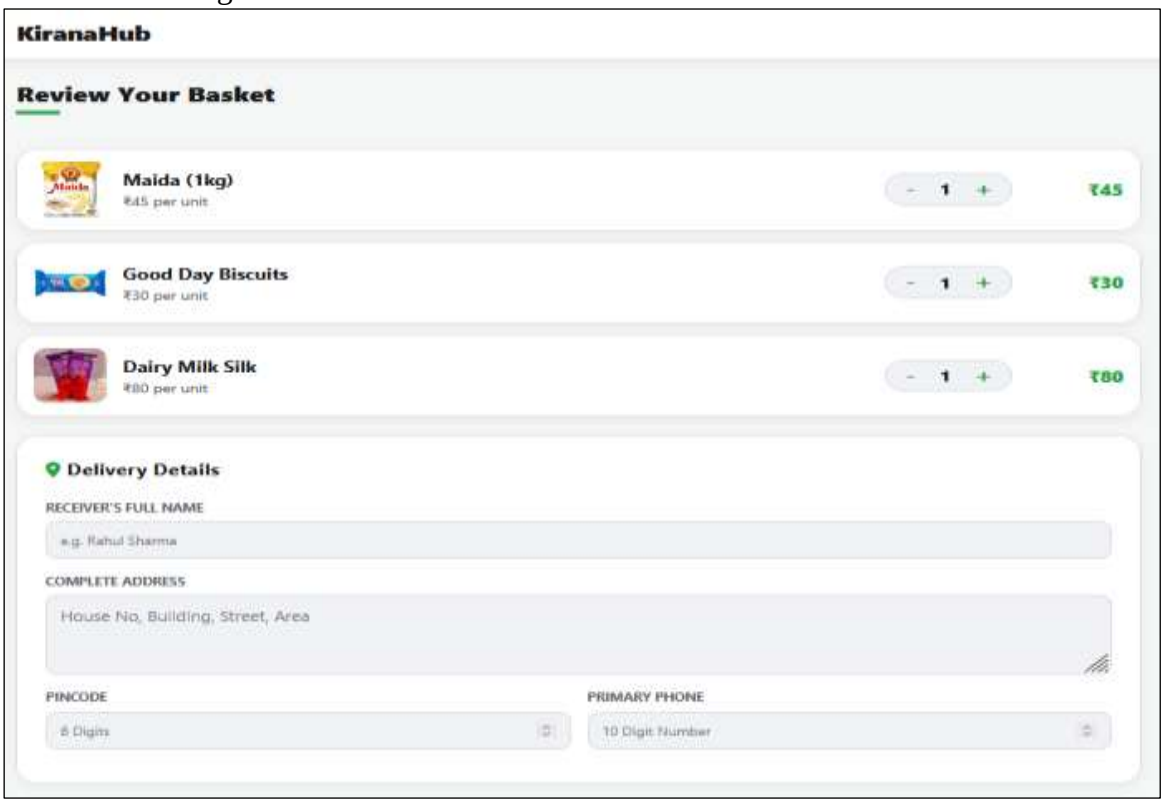

Figure 6.16 Shopping Basket Page – Customer

4. Product Review– Customer:

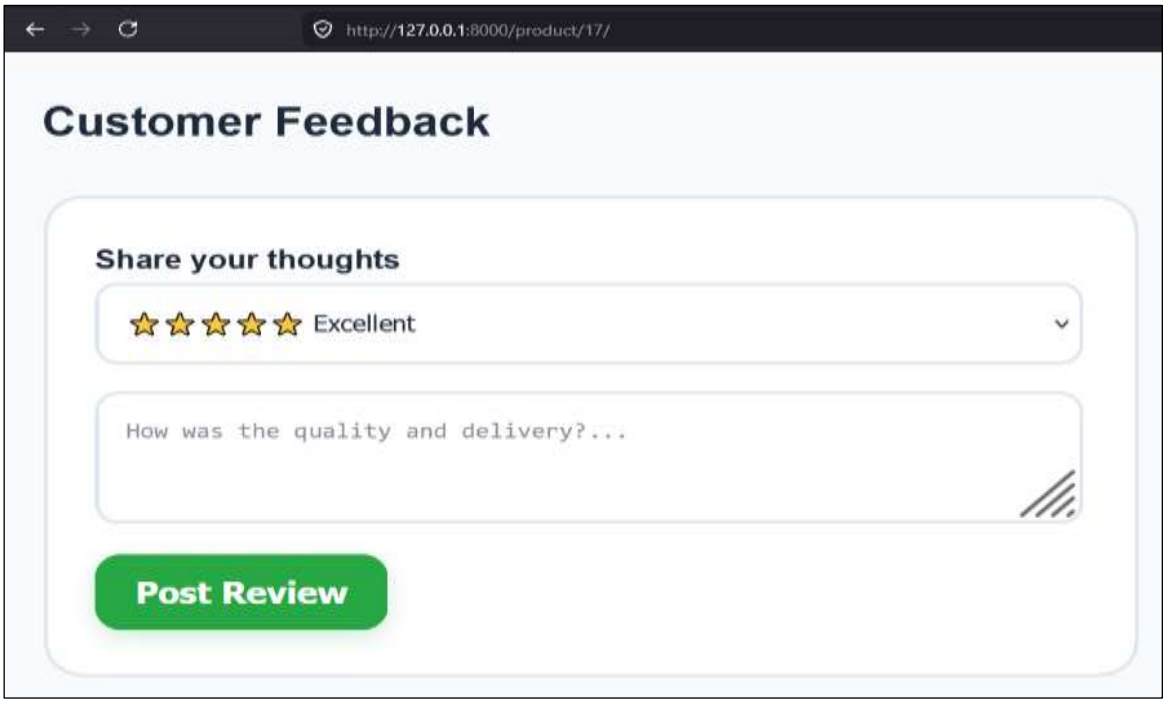

Figure 6.17 Product Review– Customer

5. User Wishlist -Customer:

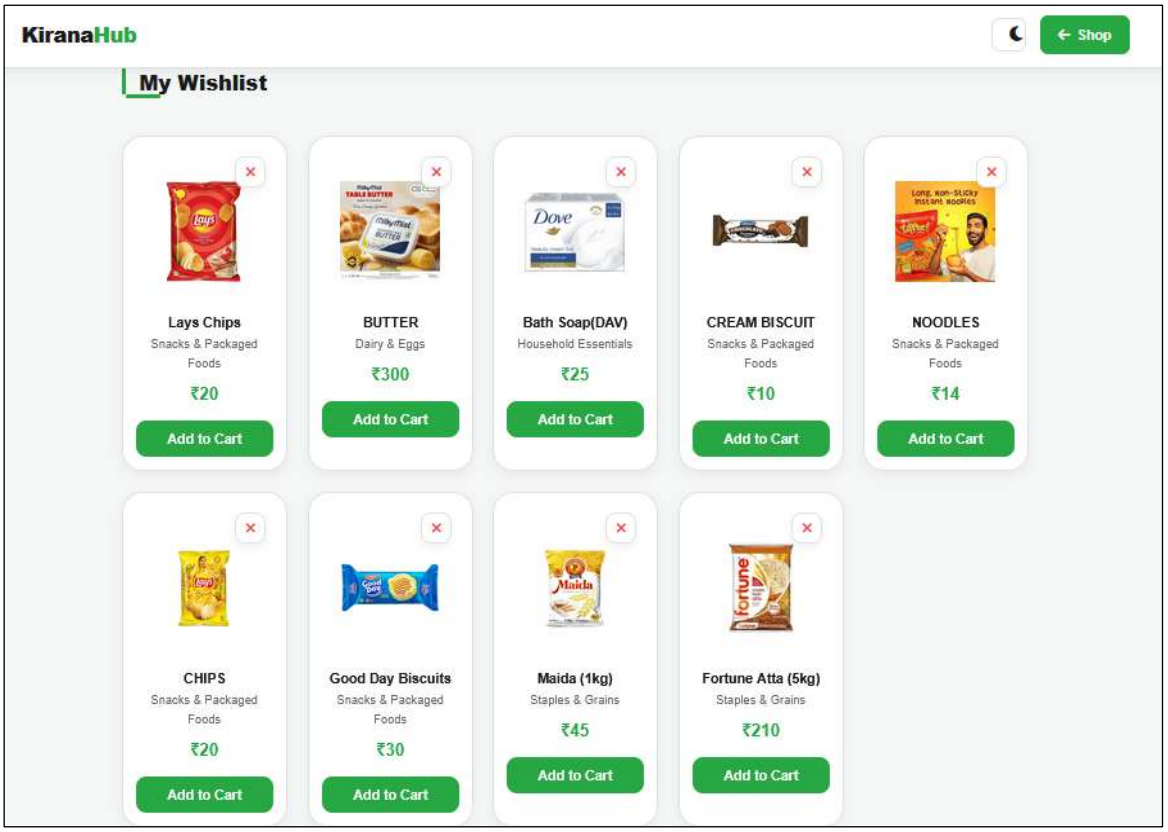

Figure 6.18 User Wishlist -Customer

6. User Profile Page – Customer:

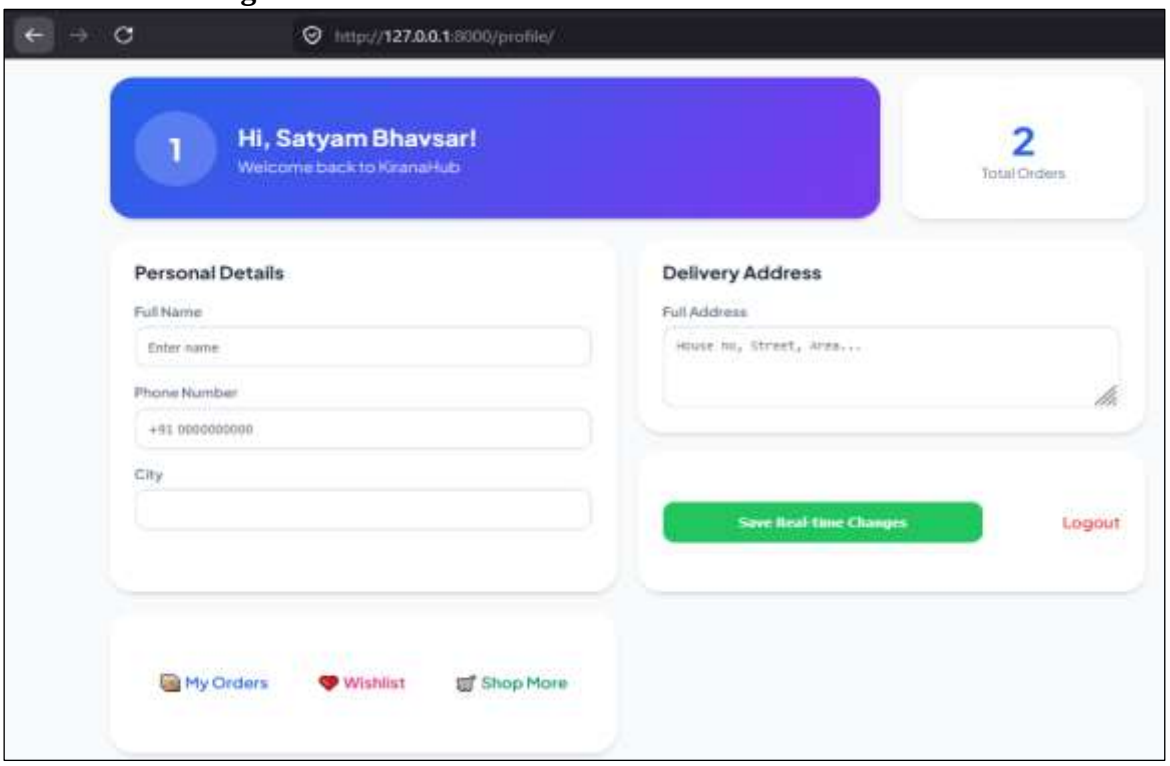

Figure 6.19 User Profile Page – Customer

-The profile page shows a welcome message with the user's name and total orders.

-Users can update personal details and delivery address with a real-time save option.

7. Help & Center – Customer:

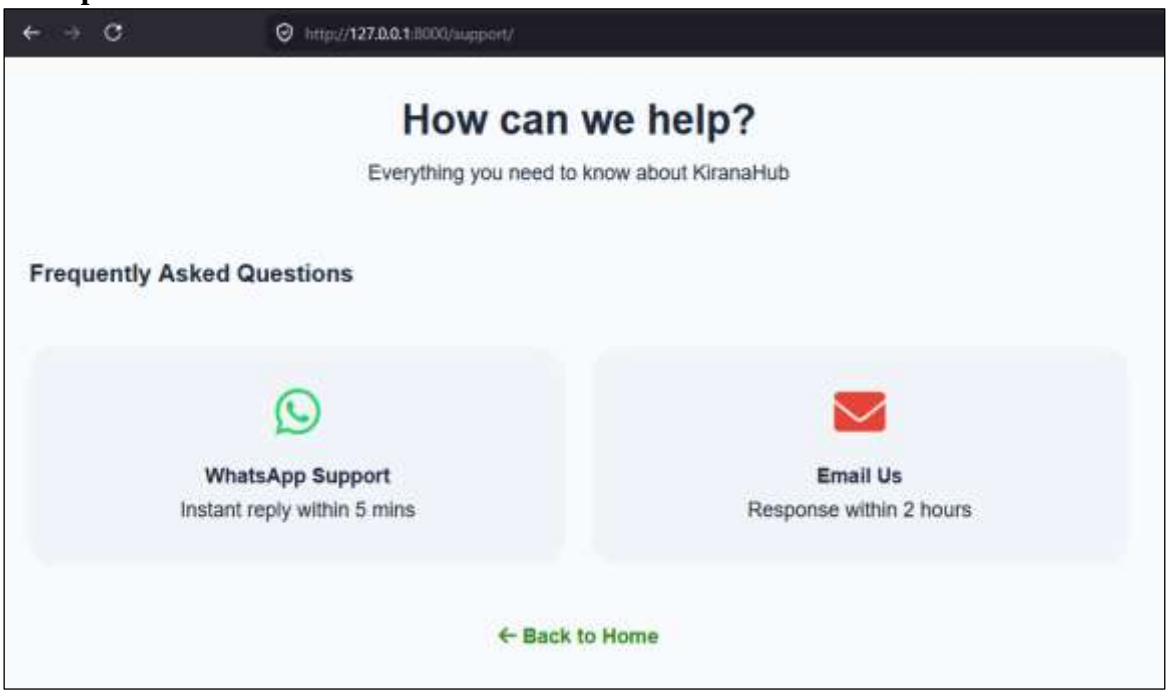

Figure 6.20 Help & Center – Customer

8. Payment Method – Customer:

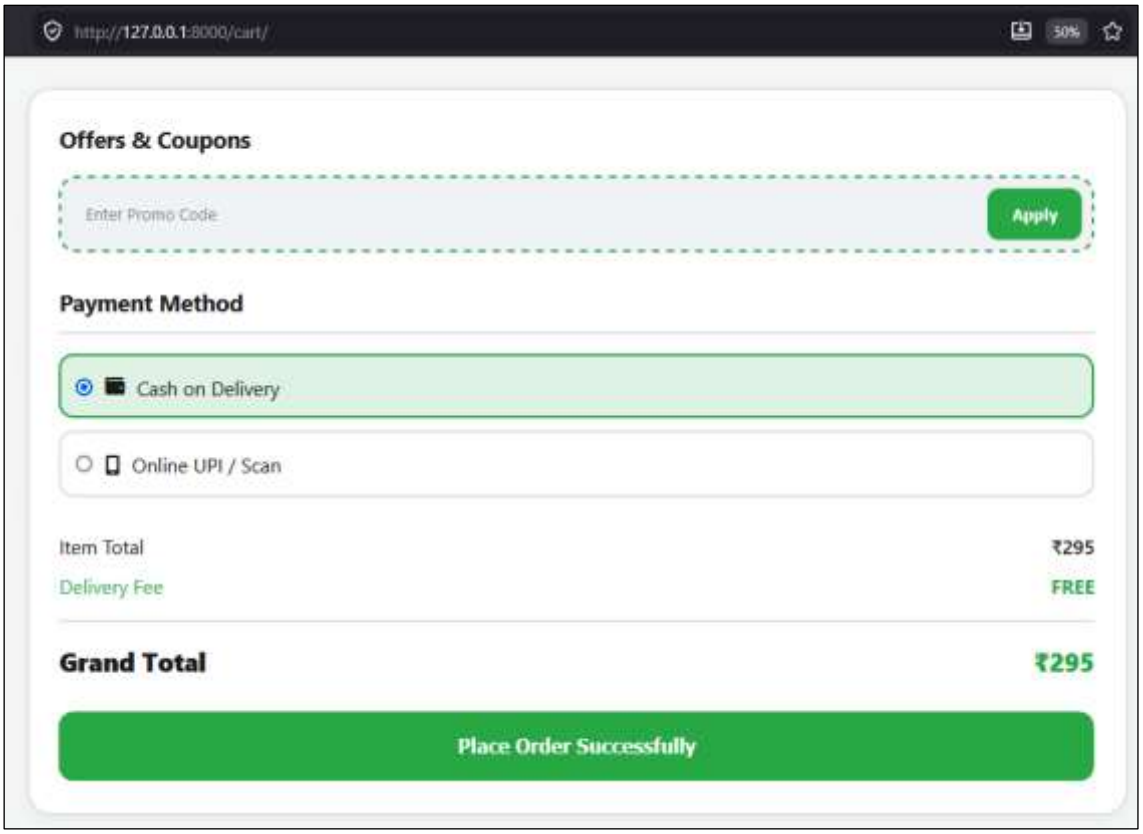

Figure 6.21 Selection of Payment Method (Cash) – Customer

- Users can apply promo codes and choose payment methods like Cash on Delivery or UPI.
- The page shows total amount with free delivery and an option to place the order.

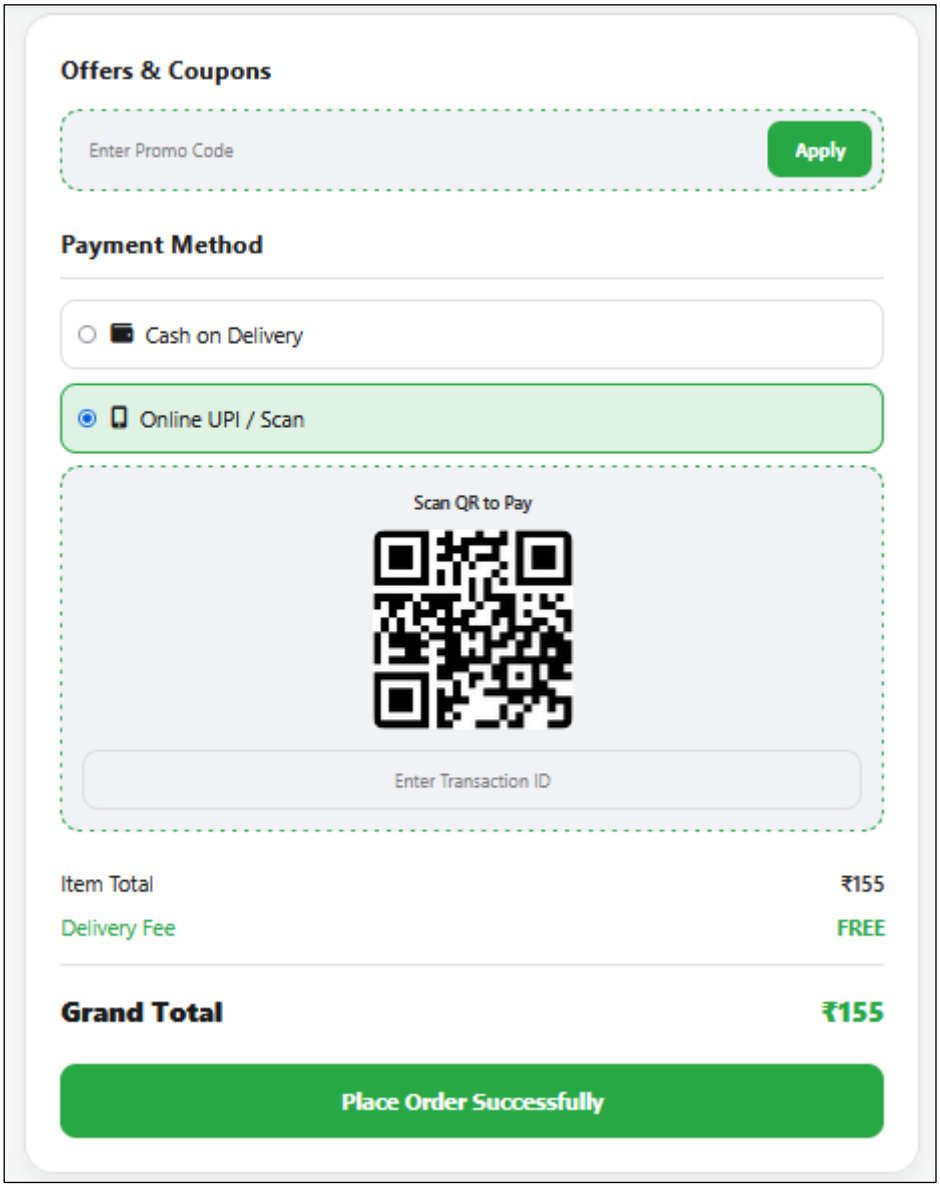

Figure 6.22 Selection of Payment Method (Online UPI/Scan)– Customer

- Order cannot be placed until payment is completed.

9. Order Tracking Page – Customer:

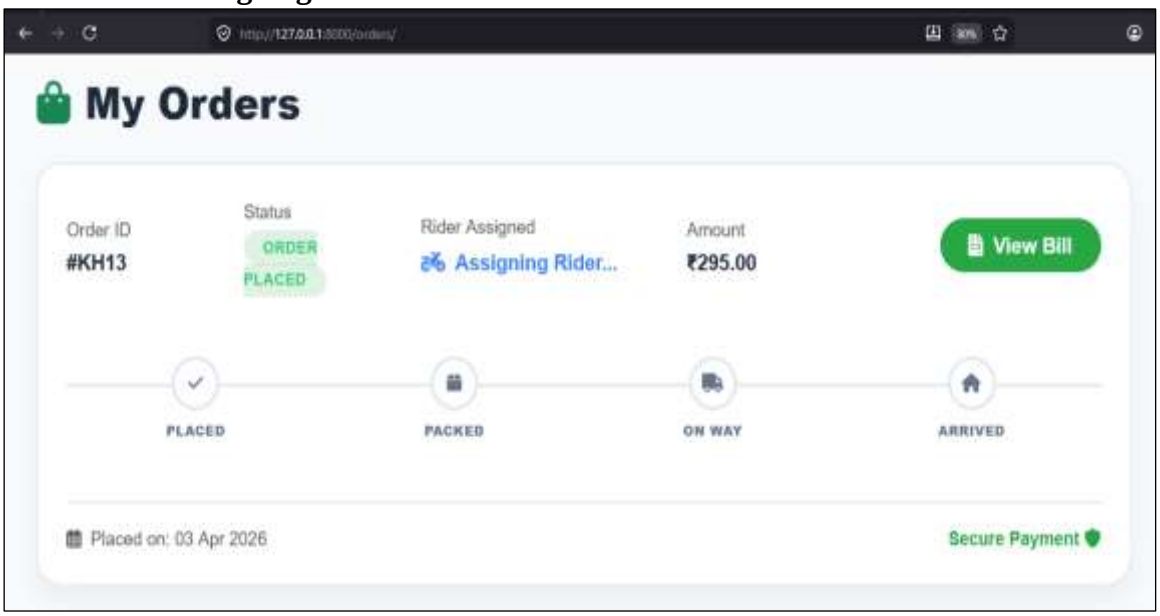

Figure 6.23 Order Tracking Page – Customer

- The page displays an order confirmation/thank you message after successful order placement.
- It provides a confirmation that the order has been received and will be processed.

10. Billing Receipt Page – Customer:

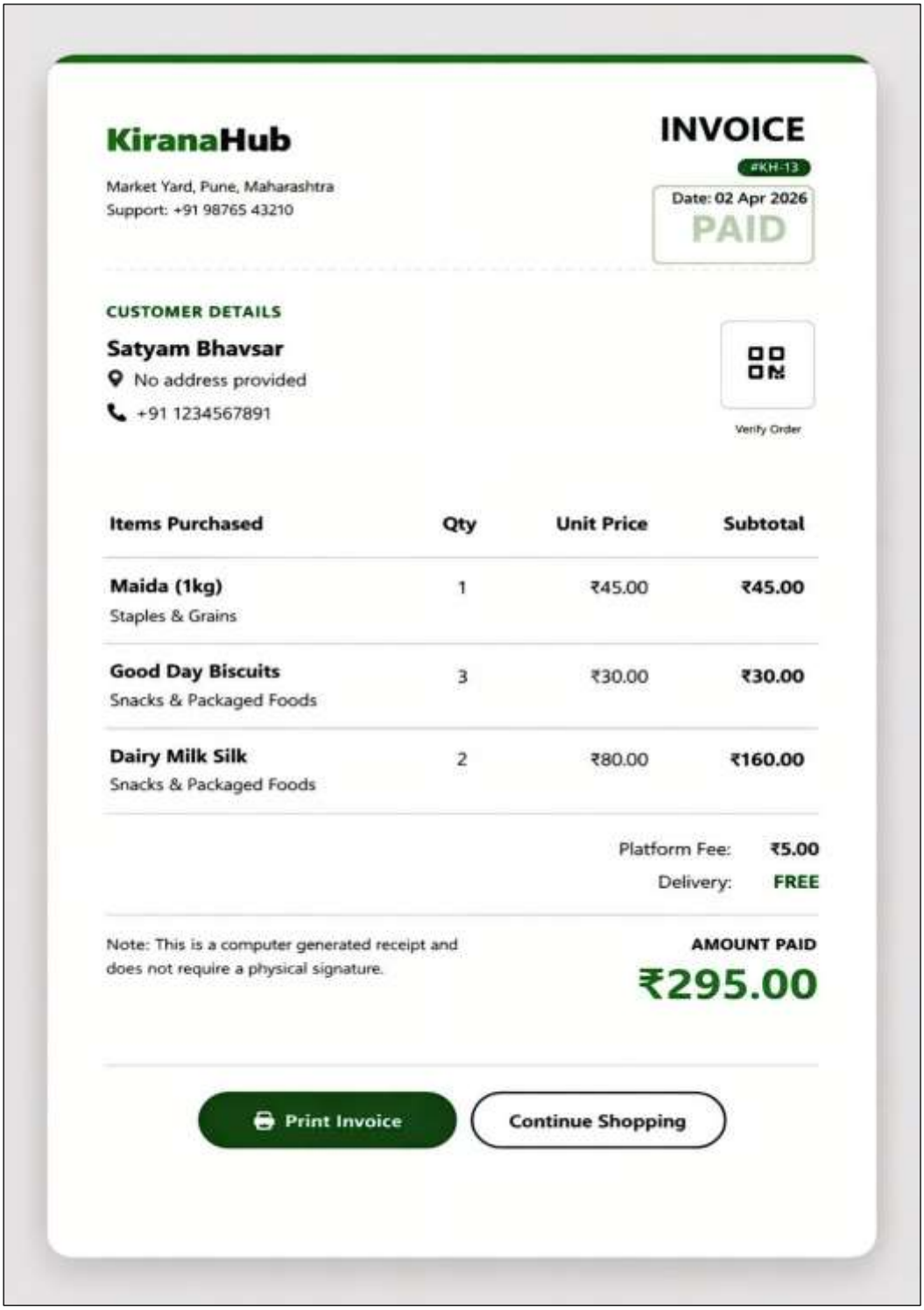

Figure 6.24 Billing Receipt Page – Customer

- The page displays a bill (invoice) with item details, quantity, and pricing.
- It shows the total amount paid along with payment status.

6.2 Application of Digital GroceryHub System

The Digital GroceryHub System is a convenient platform designed to simplify online grocery shopping for customers by allowing them to browse products, add items to their cart, and place orders easily. It supports secure payment options like Cash on Delivery and UPI, ensures efficient order management, and provides real-time updates on order status and delivery. The system also helps administrators manage products, users, and orders smoothly, making the overall process organized, time-saving, and user-friendly while improving the quality of service and customer satisfaction. Additionally, it reduces manual effort and minimizes errors in order processing. It also enables better inventory control and faster delivery services.

Chapter 7

Testing

7.1 Introduction

This chapter covers the testing activities performed for the Digital GroceryHub System. Testing of this system is critical because it serves as the platform for managing online grocery orders, product listings, user interactions for all users including customers and administrators. If any module fails or behaves incorrectly, the system may not function reliably and may affect order processing and user experience.

The objective of this testing phase is to verify that all functionalities work correctly across different modules. The system should allow valid users to register and log in successfully, browse products, add items to cart, place orders, and track order status. It must also reject unauthorized access and invalid inputs, and ensure proper validation of all required fields.

Testing was structured to cover both positive scenarios (successful login, product selection, cart operations, and order placement) and negative scenarios (wrong credentials, empty fields, invalid inputs, and unauthorized access).

Testing was performed using two tools:

- **Selenium IDE:** Record-and-playback based UI test automation tool used for quick scenario capture and regression checks.
- **Selenium WebDriver:** Scripted browser automation tool used for repeatable, assertion-based test execution with evidence-level results.

Out of scope for this chapter:

- Login module testing.
- Grocery Management & Ordering System

7.2 Test Plan

This test plan describes the strategy, scope, objectives, schedule, and deliverables for testing the Login Module of Digital GroceryHub System. The goal is to ensure that the login functionality works correctly, securely, and allows only authorized users to access the system.

Test Plan: (Digital GroceryHub System – Login Page)

Version: 1.0

Test Plan ID: TP_LOG_01

1. Introduction

This test plan verifies that the Login Page of the Digital GroceryHub System works correctly. It ensures that registered users can log in successfully using valid username and password, and invalid inputs are handled properly with appropriate error messages.It also validates input fields, password masking, and authentication process to ensure system security. The system must prevent unauthorized access and allow only valid users to enter the application. This testing ensures reliability, proper validation, and a smooth user experience on the login page.

2. Test Items

- Login form UI elements – username field, password field, login button, error messages
- Username and Password validation
- Password masking (•••••)
- Login authentication process
- Error message display for invalid or empty inputs
- Successful login confirmation

3. Features to be Tested

- Valid login with correct credentials
- Invalid login with wrong username/password
- Empty field validation
- Password masking functionality
- Error message validation
- Role-based login (Admin/User)
- Session management after login

4. Features Not to be Tested

- Registration module
- Payment system
- Order management

- Cart functionality
- Product management
- Reports and analytics

5. Approach

- Functional testing to verify valid and invalid login
- Positive and negative test cases execution
- Input validation testing for username and password
- Security testing for unauthorized access and SQL injection
- Manual testing and Selenium automation

6. Pass / Fail Criteria

- All high-priority test cases must pass successfully
- Minimum 95%–99% requirement coverage should be achieved
- System should not have any critical defects
- Expected results should match actual results
- All major defects must be fixed and retested

7. Suspension Criteria and Resumption Requirements

- Testing will be stopped if critical login issues occur
- Testing will resume after bug fixing and system stabilization

8. Test Deliverables

- Test plan document
- Test cases document
- Defect report
- Test summary report

9. Test Tasks

- Prepare test plan and test cases
- Execute login test cases
- Identify and log defects
- Retest fixed defects
- Prepare final test report

10. Test Environment

- Operating System: Windows 11
- Browser: Google Chrome, Microsoft Edge
- Frontend: HTML / CSS / JavaScript
- Backend: Java / Node.js / PHP
- Database: MySQL / MongoDB
- Server: Localhost / Test Server
- Tools: Selenium IDE, Postman
- Network: Internet connection

11. Responsibilities

- QA Lead: Review test documents
- Test Engineer: Execute tests and log defects
- Developer: Fix defects
- Project Manager: Approve release

12. Staffing and Training

Requires knowledge of login module functionality, testing techniques, and Selenium tools.

13. Scheduling

Project deadline: 20 March 2026, 12:00 PM

14. Risks

- Unauthorized access due to weak validation
- Incorrect error message handling
- Browser compatibility issues
- Server downtime

15. Approvals

Project Guide, Team Member

Test Cases :-

Table 4 Test Cases for Login Page

Test Case ID	Test Scenario	Test Data	Expected Result	Actual Result	Status
TC_L_01	Login with valid username & password	Username: Customer Password: Satyam@123	User should login successfully	User logged in successfully	Pass
TC_L_02	Login with Invalid password	Username: customer Password: Satyam123	Error message should display	"Invalid Password" message shown	Pass
TC_L_03	Login with Invalid Username	Username: wrong user Password: Satyam@123	Error message should display	"User not found" message shown	Pass
TC_L_04	Both fields empty	Username: "" Password: ""	Validation message should display	"All fields required" message shown	Pass
TC_L_05	Password masking	Enter password	Password should be hidden	Password shown as dots (••••)	Pass
TC_L_06	SQL Injection	‘OR’1’=’1	System should block again	Login Successful	Fail

Defect Report :-

Table 5 Defect Report (Login Page)

Defect ID:	DR_LOG_001
Project Name:	Digital GroceryHub System
Product:	Web Application
Release Version:	v1.0
Module:	User Login Module
Detected Build Version:	v1.1
Summary:	System allow login with invalid credentials
Description:	While executing the login module test scenarios, it was observed that the system does not validate user credentials properly. When the user enters invalid username or password multiple times and clicks on the login button, the system allows access without showing any error message. This may lead to unauthorized access and security risks.
Actual Result:	System allows login with invalid credentials without restriction
Expected Result:	System should validate login credentials and display error message for invalid attempts
Severity:	High (System allows unauthorized access which is a major security risk)
Priority:	High (Issue affects system security and needs immediate fix)
Reported By:	Test Engineer
Status:	Open

7.3 Test Plan

Test Plan: (Grocery Management & Ordering System)

Version: 1.0

Test Plan ID: TP_GRO_01

1. Introduction

This test plan verifies that the Grocery Management & Ordering System works correctly. It ensures that users can browse products, add items to cart, place orders, and manage data efficiently using valid inputs, while invalid inputs are handled properly with appropriate error messages.It also validates system functionalities like product management, order processing, and user interactions to ensure accuracy and reliability. This testing ensures smooth functioning, proper validation, and a user-friendly experience.

2. Test Items

- Product listing UI elements – product name, price, quantity
- Cart functionality – add, update, remove items
- Order placement process
- Payment process (basic validation)
- Admin product management (add/update/delete)
- Error message display for invalid inputs

3. Features to be Tested

- Valid product selection and cart operations
- Order placement with valid data
- Invalid inputs (wrong quantity, empty fields)
- Product management by admin
- Payment validation
- Order tracking functionality

4. Features Not to be Tested

- Third-party payment gateway integration
- Advanced security testing
- Performance/load testing
- External API integration

5. Approach

- Functional testing of all modules (product, cart, order)
- Positive and negative test cases execution
- Input validation testing
- Manual testing and Selenium automation
- Basic security validation

6. Pass / Fail Criteria

- All high-priority test cases must pass successfully
- Minimum 95%–99% requirement coverage should be achieved
- System should not have any critical defects
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- All major defects must be fixed and retested

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- Network: Internet connection

11. Responsibilities

- QA Lead: Review test documents
- Test Engineer: Execute tests and log defects
- Developer: Fix defects
- Project Manager: Approve release

12. Staffing and Training

Requires knowledge of system modules, testing techniques, and automation tools like Selenium.

13. Scheduling

Project deadline: 20 March 2026, 12:00 PM

14. Risks

- Incorrect order processing
- Payment validation issues
- Data inconsistency
- Server downtime

15. Approvals

Project Guide, Team Member

- Grocery Management & Ordering System

Table 6 Test Cases for Grocery Management & Ordering System

Test Case ID	Test Scenario	Test Data	Expected Result	Actual Result	Status
TC_G_01	Add product with valid details	Product: Rice Price: 50 Qty: 10	Product should be added successfully	Product added successfully	Pass
TC_G_02	Add product with empty fields	Product: ""	Error message should display	"All fields required" shown	Pass
TC_G_03	Add product with negative price	Product: Sugar Price: -20 Qty: 5	System should reject invalid price	Product added with negative price	Fail
TC_G_04	Update product price	Change price from 50 to 60	Product should update	Product updated successfully	Pass
TC_G_05	Delete product	Select product & delete	Product should be removed	Product deleted successfully	Pass
TC_G_06	Add to cart with valid quantity	Product: Rice Qty: 2	Product should be added to cart	Product added to cart	Pass
TC_G_07	Payment with valid amount	Total: 200	Payment should be successful	Payment failed	Fail

Defect Report :-

Table 7 Defect Report (Grocery Management & Ordering System)

Defect ID	DR_GRO_001
Project Name	Digital GroceryHub System
Product	Web Application
Release Version	v1.0
Module	Product Management
Detected Build Version	v1.1
Summary	System allows product with negative price
Description	While testing the product module, it was observed that the system accepts negative price values. When user enters a product price as negative (e.g., -20) and submits, the product is added successfully without any validation. This may lead to incorrect billing and data inconsistency.
Actual Result	Product added with negative price
Expected Result	System should reject negative price and display validation error
Severity	High (Affects billing and data accuracy)
Priority	High (Needs immediate fix)
Reported By	Test Engineer
Status	Open

Chapter 8

Conclusion and Future Scope

1.1 Conclusion:

The Digital GroceryHub System is an efficient and user-friendly web-based application that automates the entire process of browsing, selecting, and ordering household essentials. By replacing traditional manual shopping methods with a digital solution, the system improves accuracy, reduces the time required for purchasing daily items, and enhances overall customer convenience.

The application provides real-time product availability, transparent pricing, and secure handling of user data such as addresses and order history. It helps in efficient order processing and ensures smooth management of grocery deliveries. Overall, the system offers a reliable and modern solution for managing daily grocery needs effectively.

Furthermore, the system minimizes human errors and reduces dependency on manual work, which improves operational efficiency. It also ensures better organization of data related to users, products, and orders, making the system easy to manage and maintain. The simple and interactive user interface makes it accessible to a wide range of users, even those with basic technical knowledge.

In conclusion, the Digital GroceryHub System successfully fulfills its objective of providing a convenient, secure, and efficient platform for online grocery shopping, making it a valuable solution in today’s fast-paced lifestyle.

1.2 Future Scope:

The Digital GroceryHub System can be further enhanced by integrating advanced technologies such as AI-based recommendation systems, which can suggest products to users based on their purchase history and preferences. This will improve user experience and make shopping more personalized and efficient. Additionally, implementing a smart search and filtering mechanism can help users quickly find products, saving time and effort.

The system can also be improved by introducing advanced inventory and order management features, such as automatic stock updates, low-stock alerts, and real-time order tracking. Integration of GPS-based tracking can allow users to monitor the exact status and location of their orders. Adding multiple secure payment options like UPI, digital wallets, and net banking will further enhance convenience and flexibility for users.

In the future, the system can be expanded into a scalable and fully-featured e-commerce platform by supporting multiple locations and a larger user base. Security can be strengthened using OTP-based authentication and data encryption techniques to protect user data. Moreover, developing a dedicated mobile application can increase accessibility and user engagement, making the system more efficient, secure, and widely usable.

Chapter 9

Reference and Bibliography

1. Reference:

- **Books**

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- **ACADEMIC & RESEARCH REFERENCES**

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2. E-Commerce Grocery Management System

   https://www.researchgate.net/publication/Ecommerce_Grocery_System

3. Web-Based Grocery Store System

   https://www.researchgate.net/publication/Web_Based_Grocery_System

4. Digital Retail and Online Shopping System

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2. Bibliography:

The development of the Digital GroceryHub System is supported by various online platforms and research resources. Websites such as BigBasket and Blinkit provided insights into real-time grocery delivery systems and user-friendly interfaces. The grocery section of Amazon and Flipkart helped in understanding product management and online transaction processes. Learning platforms like GeeksforGeeks, W3Schools, and TutorialsPoint were used for coding guidance and implementation of various functionalities. Problem-solving and debugging support was taken from Stack Overflow. Research materials and academic references were collected from ResearchGate. Additionally, JioMart and Reliance Retail provided an understanding of modern retail systems. Backend and database support concepts were studied using Google Firebase and SQLite. Official development guidelines were referred from Android Developers. Version control and project resources were managed with the help of GitHub. These sources collectively contributed to the successful design and development of the proposed system.